

CHAPTER 5

MAINTENANCE



CHAPTER 5

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SECTION 5-1

START-UP CHECKLIST

Your **Peerless** mixer is designed to give quality service beginning at start-up. However, as with any new piece of equipment, there are certain checks you should make before and during the first days of operation to ensure smooth and efficient operation of your mixer.

Table 5-1-1, Start-Up Checklist, is a quick reference list of recommended areas to be checked at start-up of the mixer and after one week of operation.

SECTION 5-1

START-UP CHECKLIST TABLE 5-1-1

SUGGESTED AREA		PROCEDURE
Agitator		Check for loose pins. See Section 5-8 .
		Check for side-to-side motion. See Section 5-9 .
Left-Hand Agitator Shaft Bearings		Check for undue noise, vibration, excessive heat. See Section 5-9 .
		Check for loose bolts; retorque all bolts. See Section 5-9 .
Bowl Mounting		Retorque all bolts. See Section 5-10 .
		Inspect cam rollers for binding, proper lubrication. See Section 5-10 .
Bowl Limit Switches		Check adjustment. See Section 5-11-13 .
Drive System	Drive Chain	Check tension, oiler. See Section 5-12-1 .
	Drive Belt (Optional)	Check tension, tracking. See Section 5-12-2 .
	Drive Motor	Check for undue noise, excessive heat. See Section 5-12-3 .
	Right-Hand Gear Case	Check for oil level, oil leaks, bearing or gear noise. See Section 5-12-6 .
		Check for loose bolts. See Section 5-12-6 .
	Gear Reducer	Check for undue noise, excessive heat. See Section 5-11-5 .
		Retorque all bolts. See Section 5-11-5 .
	Motor Coupling	Check alignment. See Section 5-12-10 .
Electrical System		Check all terminal screws and lugs to be sure they are tight.
Hydraulic Tilt System		Check for leaks. See Section 5-11 .
Pneumatic System		Check for leaks. See Section 5-12-14 .
Refrigeration System		Check for leaks. See Section 5-15 .
		Check for twisting or binding of hose. See Section 5-15 .

SECTION 5-2

PREVENTIVE MAINTENANCE

The following Preventive Maintenance Schedule is a summary of the recommended maintenance required for your **Peerless** mixer. Each item is explained in detail after **Chart 5-2-1, Recommended Preventive Maintenance Schedule**. Sometimes personal experience dictates an inspection schedule different from our recommendation. Follow all safety and preventive maintenance guidelines when working on the mixer.

Your **Peerless** mixer has been designed and built with a minimum of lubrication requirements. However, it is *important* that you develop and enforce a thorough, periodic maintenance program that covers your mixer's lubrication needs.

⚠ DANGER

High electrical voltage is present in remote starter enclosure. Electrical shock can cause serious injury or death. Disconnect electrical power and **lockout** before servicing or cleaning mixer.

⚠ DANGER

Before servicing, repairing, or cleaning mixer, follow **LockOut/TagOut** procedure as required by **OSHA Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

CAUTION

Before cleaning, inspecting or in any way using mixer, read and understand **Chapter 1, Safety**; and **Chapter 4, Operation, Sections 4-1, Operator Controls; 4-6, Operating Procedures; and 4-8, Operator Trouble-Shooting Guide**.

SECTION 5-2

RECOMMENDED PREVENTIVE MAINTENANCE SCHEDULE TABLE 5-2-1

MAINTENANCE AREA	PROCEDURE	FREQUENCY
Agitator	Check for loose pins.	Six Months
	Check for side-to-side motion.	Monthly
Left-Hand Agitator Shaft Bearings	Check for undue noise, vibration, excessive heat.	Monthly
Auto Central Lube System	Check grease level.	Weekly
Bowl Jacket Pressure Relief Valve	Check for blockage.	Weekly
Bowl Limit Switches	Check adjustment.	Weekly
Bowl Mounting	Check lubrication.	Monthly
	Retorque all bolts.	
	Inspect cam rollers for binding, damage or wear. Replace as needed.	
Bowl Seals	Check for cuts and tears.	Weekly
Canopy	Check flour gate system.	Monthly
	Check for air leaks.	
Belt Drive System	Check tension, wear.	Weekly
Chain Drive System	Check oil level. Check and adjust chain tension. Check sprockets and chain for wear.	Weekly
Drive Motor	Check for unusual noise, vibration.	Weekly
	Lubricate.	Six Months
EMERGENCY STOP Button	Test by pushing in with mixer running.	Weekly
Right-Hand Gear Case Assembly	Check for unusual noise, heat, vibration.	Weekly
	Check oil level, quality.	Monthly
	Drain, flush, refill reservoir.	Six Months

SECTION 5-2

RECOMMENDED PREVENTIVE MAINTENANCE SCHEDULE TABLE 5-2-1
(con't.)

MAINTENANCE AREA	PROCEDURE	FREQUENCY
Gear Reducer	Check for unusual noise, heat, vibration.	Weekly
	Check for oil leaks.	
	Check oil level, quality.	Monthly
	Drain, flush, refill reservoir.	Six Months
Hydraulic Tilt Cylinders	Check for oil leakage.	Weekly
	Manually grease rod end.	Monthly
	Check for smooth operation.	
	Check system pressure and settings.	
Hydraulic Tilt System	Check filter gauge, fluid level, fittings. Check for leaks.	Weekly
	Change fluid.	Six Months
Manual Central Lube System	Add grease to zerk.	Weekly
Motor Coupling	Check rubber element. Check set screws.	Monthly
Pneumatic System	Check filters; drain bowls. Check for air leaks	Weekly
Coolant System	Check for coolant leaks.	Monthly

SECTION 5-2**WEEKLY MAINTENANCE DETAIL 5-2-2**

The following is a detailed explanation of the Recommended Preventive Maintenance Schedule for your **Peerless** mixer. This section details, according to frequency of recommended maintenance, each item listed in **Table 5-2-1, Recommended Preventive Maintenance Schedule**. If you have any questions, call the Service Department at **Peerless Machinery Corp.**

We recommend weekly inspection and maintenance on the following areas and components of your **Peerless** mixer. Follow all safety and preventive maintenance guidelines when working on the mixer.

WEEKLY MAINTENANCE:

Auto Central Lube System (Optional).

- ✓ Check level of grease in reservoir and add if needed. See **Section 5-7, Auto Central Lube System**.

Belt Drive System (Optional).

- ✓ Check belt and pulleys for wear. Check belt tension. See **Section 5-12-2, Drive Belt System**.

Bowl Jacket Pressure Relief Valve.

- ✓ Be sure valve is not plugged with dirt or dough. Clean as needed.

Bowl Limit Switches.

- ✓ Observe bowl stops and **bowl limit switches** during operation. Make sure switches are set so bowl will not hit bowl-stops hard. If the bowl hits bowl-stops hard, or not at all, adjust limit switches. See **Section 5-11-13**.

Bowl Seals.

- ✓ Check front and rear rubber bowl seals for cuts and tears.

SECTION 5-2

WEEKLY MAINTENANCE (con't.):

Chain Drive System.

- ✓ Check oil level in drip oiler. Fill one-pint reservoir on automatic drive-chain oiling system. See **Section 5-12-1**.
- ✓ Inspect chain for wear and proper lubrication. Adjust tension on **chain** if slack exceeds measurements in **Figure 5-12-3, Roller Chain Tension**. See **Section 5-12-4, Roller Chain Wear**.
- ✓ Be sure **chain** is moist with lubricant but not dripping or slinging oil.
- ✓ Check **sprockets** for wear.

NOTE: **Automatic Electric Oiler** is factory preset at a feed rate of approximately ten drops per minute at sight glass. Increase or decrease preset as needed depending on usage of mixer.

NOTE: **Automatic Electric Oiler** only lubricates while mixer is running in low speed. Lubricating in high speed causes excessive oil sling and waste.

Drive Motor.

- ✓ Listen for unusual **motor** noise. Feel **motor** housing for undue vibration while mixer runs loaded and unloaded. See **Section 5-12-3**.

Emergency STOP Button.

- ✓ Test by pushing in with mixer running.

Right-Hand Gear Case Assembly.

- ✓ Listen to **right-hand gear case assembly** for any unusual noises. Feel for unusual heat or excessive vibration while mixer runs, loaded and unloaded. Check for oil leakage around unit. See **Section 5-12-6**.

SECTION 5-2

WEEKLY MAINTENANCE (con't.):

Gear Reducer.

- ✓ Listen to **reducer** for any unusual noises. Feel for unusual heat or excessive vibration while mixer runs, loaded and unloaded. Check for oil leakage around unit. See **Section 5-11-5**.

NOTE: Normal operating temperatures of **gear reducer** should not exceed 180° F on surface of housing or 200° F internal oil temperature.

HydraulicTilt Cylinder.

- ✓ Check for oil leakage at all fittings and around piston rod.

Hydraulic Tilt System.

- ✓ Check filter gauge while bowl is moving. Change filter when pointer reaches yellow or red area.
- ✓ Check level of hydraulic fluid in reservoir. If oil is foamy or dirty, drain, flush, replace filter, and refill.
- ✓ Check all fittings for oil leaks. See **Section 5-10**.

Manual Central Lube System.

- ✓ Add grease to zerk to distribute lubricant through system to trunnion rollers, bearings, and idler. See **Section 5-6**.

Pneumatic System.

- ✓ Check **pneumatic filter-regulator** for plugged or dirty filters; replace as needed. Drain filter bowls as needed. See **Section 5-14**.
- ✓ Check **pneumatic system** for air leaks.

⚠ DANGER Before servicing, repairing, or cleaning mixer, follow **LockOut/TagOut** procedure as required by **OSHA Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

SECTION 5-2**MONTHLY MAINTENANCE DETAIL 5-2-3**

Peerless recommends the following areas and components of your mixer be inspected and maintained on a monthly basis. Follow all safety and preventive maintenance guidelines when working on the mixer.

MONTHLY MAINTENANCE:

- | | |
|---|---|
| Agitator. | ✓ Check for side-to-side motion. See Section 5-9 . |
| Left-Hand Agitator Shaft Bearings. | ✓ Listen to agitator shaft bearings running under load for any unusual noise or vibration. |
| | ✓ Check for hotter than normal temperatures of bearing head, shaft and covers. |
| Bowl Mounting. | ✓ Retorque all bolts. |
| | ✓ Inspect all trunnion rollers for binding, damage or wear. Replace as needed. See Section 5-10 . |
| | ✓ Check for proper lubrication. |
| Canopy. | ✓ Check flour gate system for smooth motion. |
| | ✓ Check for air line leaks. |
| Gear Reducer. | ✓ Check oil level and quality. Remove red oil-level plug. Change contaminated, burned or waxed oil; flush out reservoir if necessary. See Section 5-12-5 . |

SECTION 5-2

MONTHLY MAINTENANCE *(con't.)*:

Hydraulic Tilt System.

- ✓ Check for smooth raising and lowering, pressure settings, filter gauge. See **Section 5-11**.
- ✓ Manually grease rod end.

Motor Coupling.

- ✓ Visually check rubber element.
- ✓ Check set screws. See **Section 5-12-4**.

Coolant System.

- ✓ Check for coolant leaks. See **Chapter 7**.

SECTION 5-2**SIX MONTH MAINTENANCE DETAIL 5-2-4**

Peerless recommends the following areas and components be inspected and maintained every six months. Follow all safety and preventive maintenance guidelines when working on the mixer.

SIX-MONTH MAINTENANCE:

- | | |
|-------------------------------|---|
| Agitator. | ✓ Check for loose pins. Check for any sign of agitator-to-bowl interference. See Sections 5-8 and 5-9 . |
| Drive Motor. | ✓ Lubricate. Follow steps in Section 5-11-3 . |
| Hydraulic Tilt System. | ✓ Change hydraulic fluid in reservoir. See Section 5-11 . |
| Gear Reducer. | ✓ Drain oil, flush and refill reservoir. See Section 5-12-5 . |

SECTION 5-2

RECOMMENDED LUBRICANTS CHART TABLE 5-2-5

We recommend the following lubricants for all **Peerless** equipment. This chart includes the oil manufacturer product codes used by **Peerless**. If substitutions are made, be sure the replacement product meets the same specifications as listed.

MIXER PART NAME	LUBRICANT NAME	PRODUCT CODE	SUS V@ 100
			ISO
			AGMA
EuroDrive Reducer	MobilGear 630	61087-30	1175
			220
			5EP
Hydraulic Power Unit	BP Energol HLP-HD-32	733001	171.6
			32
Chain Drip Oiler	Mobil Vactra 4	60051-0	1170
			220
Grease Zerks, Bearings & Bushings	Mobilith SHCPM Grease	64404-7	2400
Automatic Grease Lube			460
Drive Motor Bearings			
Grease Zerks--Edible Locations	Mystik HT-FG Grease	5635	3330
Right-Hand Gear Case	Mobil Extra HECLA	60124-5	4500
			680
			16

SECTION 5-3

SAFETY INTERLOCKS

Your **Peerless** mixer has been designed for the safest operation possible. However, the operation of heavy machinery requires a cautious approach and certain safeguards against careless or unsafe operating practices.

Peerless has provided **safety interlocks** for those mixer functions that have the potential to cause personal injury or damage to the machine if engaged carelessly or unsafely. For example, certain mixer functions require the use of both hands; i.e., to jog the agitator, you must push and hold the **JOG** and **SAFETY** buttons at the same time.

Chart 5-3-1, Operations Controls Summary, lists each operation control, tells what it does, and follows with a description of the **safety interlock** for any control that requires one. We've included the **INTERLOCKS** column here because the **Peerless** Service Department sometimes receives calls for problems that are due to safety interlocks.

OPERATION CONTROLS SUMMARY CHART 5-3-1

COMPONENT NAME AND TYPE	FUNCTION	INTERLOCKS
EMERGENCY STOP Lighted Push/Pull button	Stops mixer in an emergency.	Must be pulled out for mixer to start.
FLOUR GATE Selector Switch	Opens and closes flour gate.	Mixer will not start with gate open. Gate will not open if mixer is running. Bowl must be in full up position.
JOG Pushbutton	Jogs, or runs, agitator to discharge dough.	Must push SAFETY button also. End door, flour gate and optional ingredient door must be closed.

SECTION 5-3

COMPONENT NAME AND TYPE	FUNCTION	INTERLOCKS
LOWER Pushbutton	Lowers bowl.	Must push SAFETY button also. End door, flour gate and optional ingredient door must be closed.
MIXER STOP Pushbutton	Stops mixer run cycle.	None
RAISE Pushbutton	Raises bowl.	Must push SAFETY button also. End door, flour gate, and optional ingredient door must be closed.
RESET/POWER ON Lighted pushbutton	Resets timers. Stays lit when mixer power is on.	None
SAFETY Pushbutton	Used for two-hand operation when jogging, raising or lowering bowl.	Must be used with JOG , RAISE or LOWER pushbuttons.
START Pushbutton	Start mixer run cycle.	Mixer will not start if bowl is not full up, end doors are open, flour gate is open, optional ingredient door is open, or timers are not set.

NOTE: Always use **MIXER STOP** button to stop mixer in a non-emergency situation. Use **EMERGENCY STOP** button only in an emergency to reduce wear on master control relay coil and contacts. Test **EMERGENCY STOP** button at least once a week for proper operation.

SECTION 5-4

MAINTENANCE TROUBLE-SHOOTING GUIDE

The following table is designed to be a quick reference for trouble spots that may arise with your **Peerless** mixer. This trouble-shooting guide explains at a glance the trouble, possible causes and remedies. A list with more detailed information for each item follows **Table 5-4-1, Maintenance Trouble-Shooting Guide**.

NOTE:

- 1) To use **RAISE**, **LOWER** and **JOG** buttons, push **SAFETY** button at same time.
- 2) To operate mixer, you must pull out **EMERGENCY STOP** button.

MAINTENANCE TROUBLE-SHOOTING GUIDE TABLE 5-4-1

TROUBLE	CAUSE	REMEDY
Agitators jog when button is pushed, but will not run.	Open timer circuit.	Set timer.
	Bowl not fully closed.	Close bowl.
Agitators will not jog.	End doors open.	Close end doors.
	Ingredient door open.	Close ingredient door.
	Flour gate open.	Close flour gate.
Agitators rub bowl end.	Loose set screws in bearing adjusters.	Locate bearing adjustment screws; center agitator in bowl and retighten.
Bowl will not tilt.	Ingredient door open.	Close and latch ingredient door.
	End door open.	Close end door.
	Flour gate open.	Close flour gate.
	Low hydraulic fluid level.	Add hydraulic fluid.
	Overload relay tripped.	Reset. Replace if defective.
	Magnetic coil.	Replace.
	Loose wire.	Locate and tighten.
	Broken wire.	Locate; correct or replace.
	Defective pushbutton.	Replace.
	Motor burned out.	Remove; test, repair or replace.

SECTION 5-4

MAINTENANCE TROUBLE-SHOOTING GUIDE TABLE 5-4-1
(con't.)

TROUBLE	CAUSE	REMEDY
Drive motor runs. Agitators will not turn.	Broken coupling.	Replace broken part. Realign coupling.
	Broken drive chain or drive belt.	Repair or replace.
	Broken reducer gear or shaft.	Remove and replace defective part.
	Broken sprocket or pulley.	Replace.
Excessive noise.		Locate source by inspection. Replace malfunctioning part.
Flour gate will not open.	Bowl not closed.	Close bowl.
	No air pressure.	Check air pressure.
	Solenoid valve faulty.	Repair or replace.
Mixer will not start.	Bowl not fully closed.	Close bowl.
	Ingredient door open.	Close and latch ingredient door.
	End door open.	Close end door.
	Flour gate open.	Close flour gate.
	Timer set wrong.	Reset or enter time.
	Timer not working.	Replace.
	Open circuit breaker.	Close or reset breaker.
	Overload relay tripped.	Reset; replace if defective.
	Loose connections.	Locate and tighten.
	Defective pushbutton.	Replace.
	Starter coil.	Replace.
	Broken control wire.	Locate; correct or replace.
	Motor burned out.	Remove; repair or replace.
Refrigeration will not run. Light flashing.	Suction-line thermostat PLC input not jumpered out.	Add jumper to terminal strip.

SECTION 5-4**MAINTENANCE TROUBLE-SHOOTING GUIDE TABLE 5-4-1**
(con't.)

TROUBLE	CAUSE	REMEDY
Shaft seal leaks.	Seals dirty.	Remove, clean and relubricate.
	Improperly assembled.	Take apart and reassemble.
	Worn seals.	Replace.
Thermometer does not register.	Defective mechanical thermometer.	Replace.
	Defective digital thermometer.	Check indicator, wire and probe. Replace or repair defective part.

SECTION 5-5

REPAIR PARTS ORDERING

To make locating and ordering **Peerless** mixer parts easier, the manual contains numbered figures which call out all replaceable parts with bubble numbers. Accompanying each figure is a table that names the called-out parts in order of bubble numbers.

To locate a part, find the figure number of the drawing that contains the part. Find the part's bubble number on the figure. Refer to the table below the figure for the name of the part and the quantity used.

When ordering parts, call the **Peerless Parts Department** and give the Parts Manager the following information:

- ✓ your mixer's serial number
- ✓ the name of the part
- ✓ the part's bubble number
- ✓ the figure number the part is located on.

From this information, the Parts Manager can easily determine the part number for the part from the Customer Machine Order file kept for each mixer made.

SECTION 5-6

MANUAL CENTRAL LUBE SYSTEM

Your **Peerless** mixer comes standard with a **Manual Central Lube System**. A **Manual Central Lube System** allows maintenance personnel to lubricate five major areas of the mixer from one centrally located grease zerk fitting. See **Figure 5-6-1**. The **Manual Central Lube System** saves on maintenance time and assures efficient lubrication of your mixer.

To activate the **Manual Central Lube System**, uncap the zerk fitting mounted on the rear outside wall of the mixer or the zerk mounted on the inboard column inside the left-hand end door. For a typical application, use a standard grease gun to pump approximately 38 strokes (1.5 oz., 2.7 cu. in.) of grease into the zerk. The grease travels to a **divider valve** that distributes it proportionately to the right- and left-hand trunnion rings, the right- and left-hand bearing heads, and the drive chain idler.

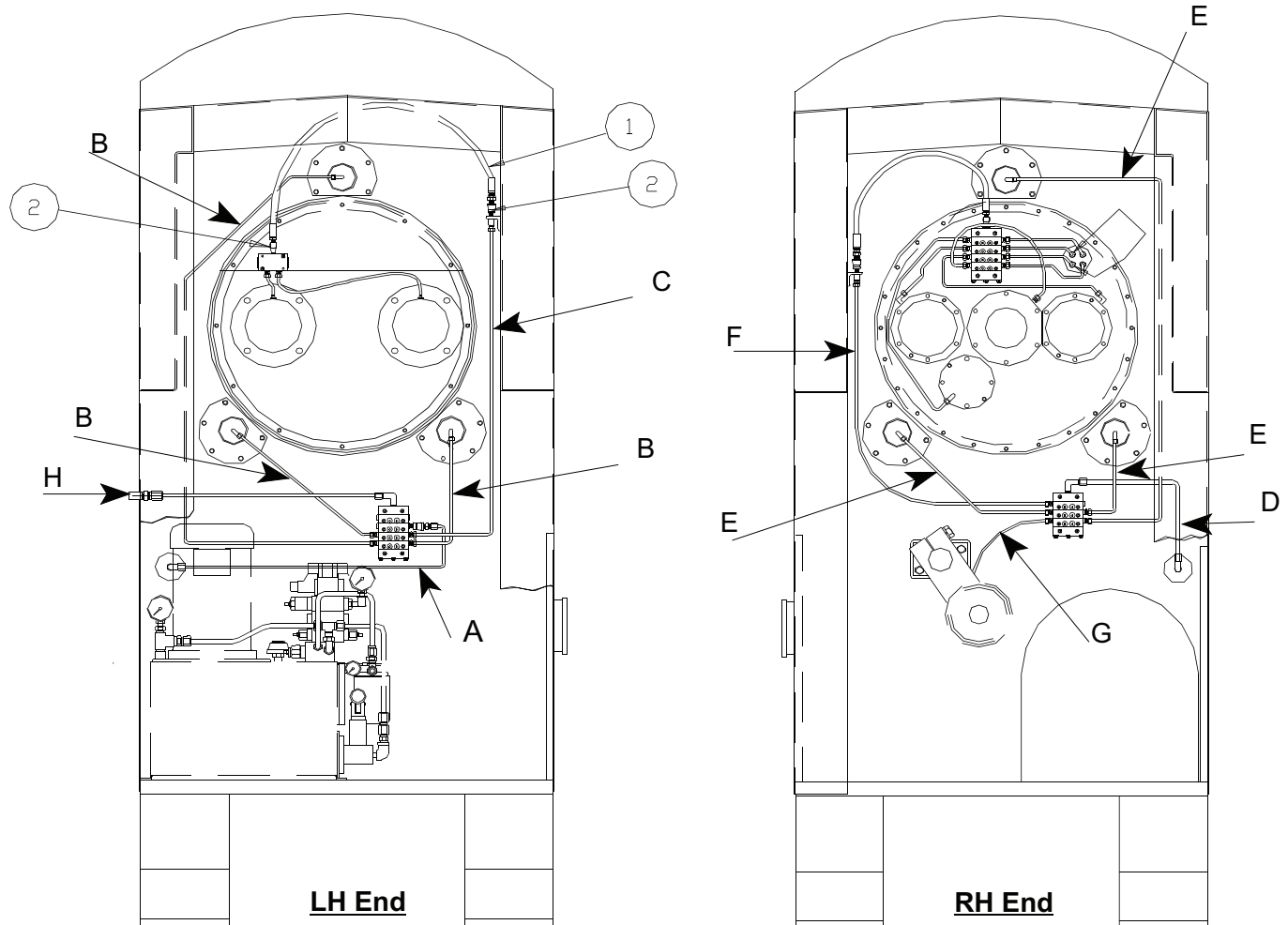
NOTE: Belt drive idlers do not require additional lubrication maintenance as provided by the **Manual Central Lube System**.

The **divider valve** is a series progressive unit consisting of modular sections that fill and distribute grease one at a time in sequence. Positive displacement metering valves in each section of the **divider valve** assure that each section is filled with the correct amount of grease and that the grease is sent to its assigned area of lubrication before the next section can be filled.

Kinked or plugged lines will keep **divider valve** sections from filling and distributing grease to the mixer's lube points. When pumping grease into the **Manual Central Lube System** with a manual grease gun, if the grease handle pumps hard or not at all, there could be a blockage in the lube system. When using a pneumatic grease gun with a gauge, a grease pressure exceeding 1500 PSI could indicate the lube system is blocked.

SECTION 5-6

MANUAL CENTRAL LUBE SYSTEM FIGURE 5-6-1



ITEM	QTY	DESCRIPTION
1	2	Lube Hose
2	2	Swivel Fitting

- A. Feed Line to RH End
- B. To LH Trunnion Rollers
- C. To LH Agitator Shaft Bearings
- D. Feed Line from LH End
- E. To RH Trunnion Rollers
- F. To RH Gear Case Bearings
- G. To Idler if Chain Drive
- H. Grease Zerk for lubrication, all points

LOCATING BLOCKAGE IN A DIVIDER VALVE SYSTEM

The following step-by-step procedure is designed to help locate blockage in the lines feeding grease into and out of the **divider valve** system. See **Figure 5-6-2, Divider Valve System**.

1. Connect the grease gun to an alternate port of the master **divider valve**.
2. Remove all alternate outlet plugs and slowly pump the grease gun.
3. If grease oozes out of the open outlets, the blockage is not in this **divider valve** section. Reinstall all alternate outlet plugs and start over at step 1 on another **divider valve** section.

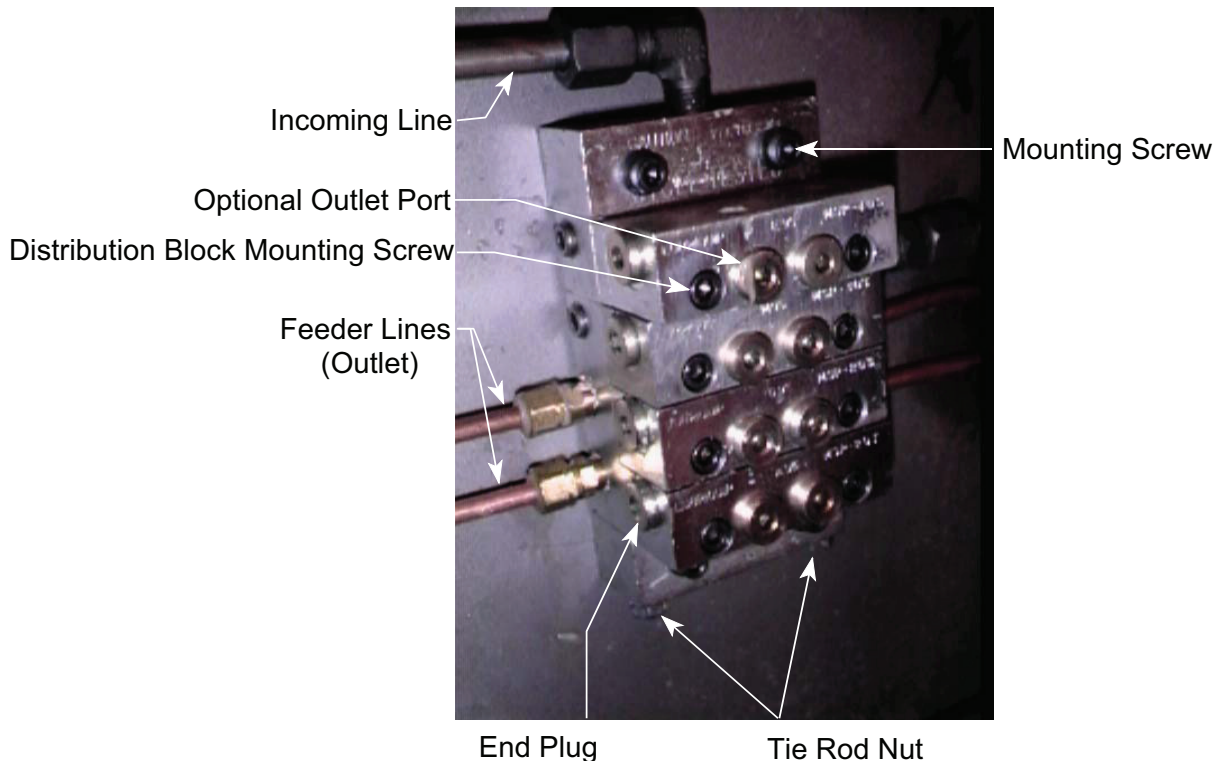
If no grease comes out of the open outlets, you have pinpointed the blockage to this **divider valve** section or its feed lines.
4. Disconnect the supply line at the secondary inlet fitting and slowly operate the grease gun to verify the location of blockage.
5. One at a time, install the grease gun in each alternate outlet of the secondary feeder. Slowly, try to pump grease into each installation point. High pressure occurs where there is blockage. Look for kinked lines or tight bearings.
6. Disassemble and clean any **divider valves** that have blockages. Before disassembling:
 - ✓ Make a sketch of the **divider valve** noting arrangement of sections, inlets, outlets and their specific uses.
 - ✓ Remove end plugs only. Try to move each piston back and forth without taking it out of its section.

If the pistons are movable and there seems to be no serious problem, replace the end plugs using a new gasket. Retorque. Retest the **divider valve** using the grease gun.

NOTE: Service and disassemble all lubricating equipment under the cleanest possible conditions.

SECTION 5-6

DIVIDER VALVE SYSTEM FIGURE 5-6-2



TO DISASSEMBLE A DIVIDER VALVE:

1. Remove tie rod nuts.
2. Remove end plug from both ends of section.
3. Remove piston from one section at a time. If piston seems to be jammed, try to remove it from the other direction. If piston is badly jammed, you might have to tap it out lightly with a brass rod.

⚠ CAUTION

Do not insert hard objects into piston bore (i.e., punches, screwdrivers, etc.). Use a brass rod and hand pressure only.

CONTAMINATION BLOCKAGE

Dirt or other problem materials can get into a **divider valve** causing blockage of the lines and failure of the **divider valve** system. Cleaning the **divider valve** is only part of the solution. To solve the problem, find and eliminate the source of contamination.

- ✓ Check system filtering method. Change filter elements if necessary.
- ✓ Check reservoir. Clean if necessary.
- ✓ Review reservoir-filling method. Use only clean lubricating tools and filtered lubricant to ensure that no foreign materials enter the reservoir during filling.

SEPARATION BLOCKAGE

Sometimes a hard wax or soap-like material is found in sections of the **divider valve** causing system failure. These deposits are signs of grease separation which occurs when oil is squeezed from the grease at normal operating pressure, leaving the grease thickener behind in the **divider valve**. Cleaning the **divider valve** is only part of the solution. Consult your lubricant supplier for recommendations on alternate lubricants.

CLEANING THE DIVIDER VALVE:

1. Take the **divider valve** system apart. Clean each section with solvent until all lubricant has been removed.
2. Dry and blow out all ports thoroughly. Use a small wire probe to make sure all ports are clean and open. Check the cylinder bore and piston for scratches, score marks or other damage.

NOTE: If either the piston or the cylinder bore is damaged, replace that section of the **divider valve**.

SECTION 5-6

3. If the **divider valve** section appears in good condition or repair is complete, reassemble making sure the piston slides smoothly but snugly inside the cylinder bore.

Because each piston is fitted to its bore for exact clearance, *be sure to reinstall each piston into the section it was taken from.*

4. Repeat cleaning and inspection for all sections. Reassemble the **divider valve**.

SECTION 5-7

AUTO CENTRAL LUBE SYSTEM

Peerless can supply your mixer with an optional **Auto Central Lube System**. See **Chapter 2, Options Checklist**. An **Auto Central Lube System** operates on the same principle as the **Manual Central Lube System** to lubricate five major areas of the mixer. See **Figure 5-7-1**. However, the **Auto Central Lube System** operates automatically using a pneumatic grease pump and a grease reservoir saving on maintenance time and assuring efficient lubrication of your mixer.

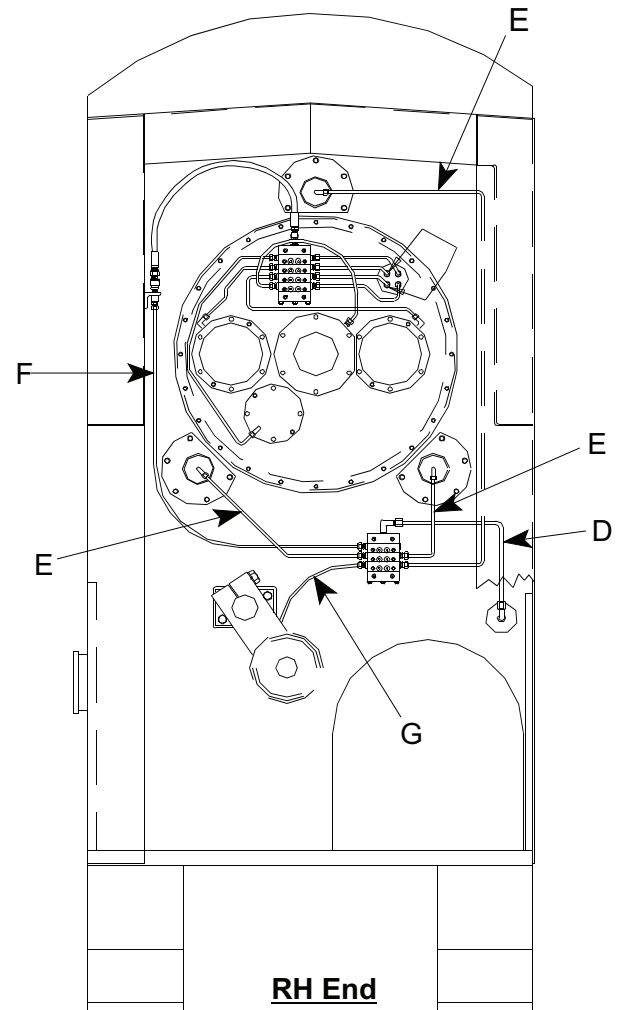
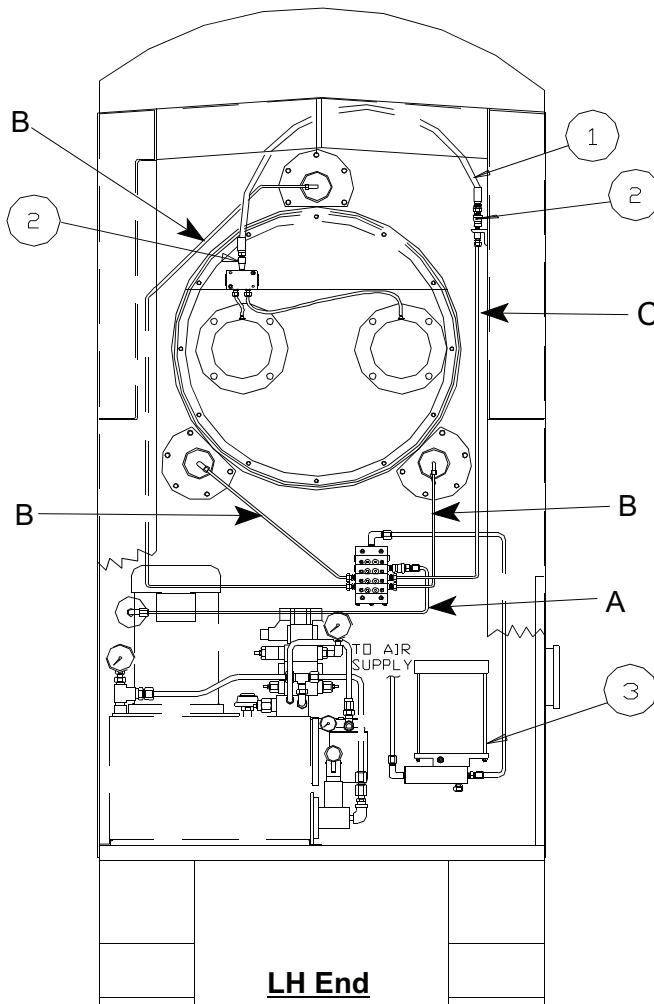
In a typical application, every eight hours that power is on to the mixer, grease is automatically pumped into the **Auto Central Lube System**. The **divider valves** distribute grease proportionately to the right- and left-hand trunnion rings, the right- and left-hand bearing heads, and the drive chain idler.

NOTE: Belt drive idlers do not require additional lubrication maintenance as provided by the **Auto Central Lube System**.

A pneumatic pump operates the **Auto Central Lube System**. A spring at the top of the grease reservoir constantly presses the lubricant down, forcing it through the **flapper valve** into the smaller **lube chamber** inside the pump. At the predetermined intervals, the **PLC** activates the solenoid valve to force air into the **air chamber cavity** at the bottom of the pump, raising the **piston**. The rising **piston** pushes the grease upward in the **lube chamber** against the closed **flapper valve**, forcing it to escape through the **lube outlet port**. From there, the grease travels to the **divider valve** to be distributed to the designated areas. See **Figure 5-7-2, Pneumatic Pump Operation**.

SECTION 5-7

AUTO CENTRAL LUBE SYSTEM FIGURE 5-7-1

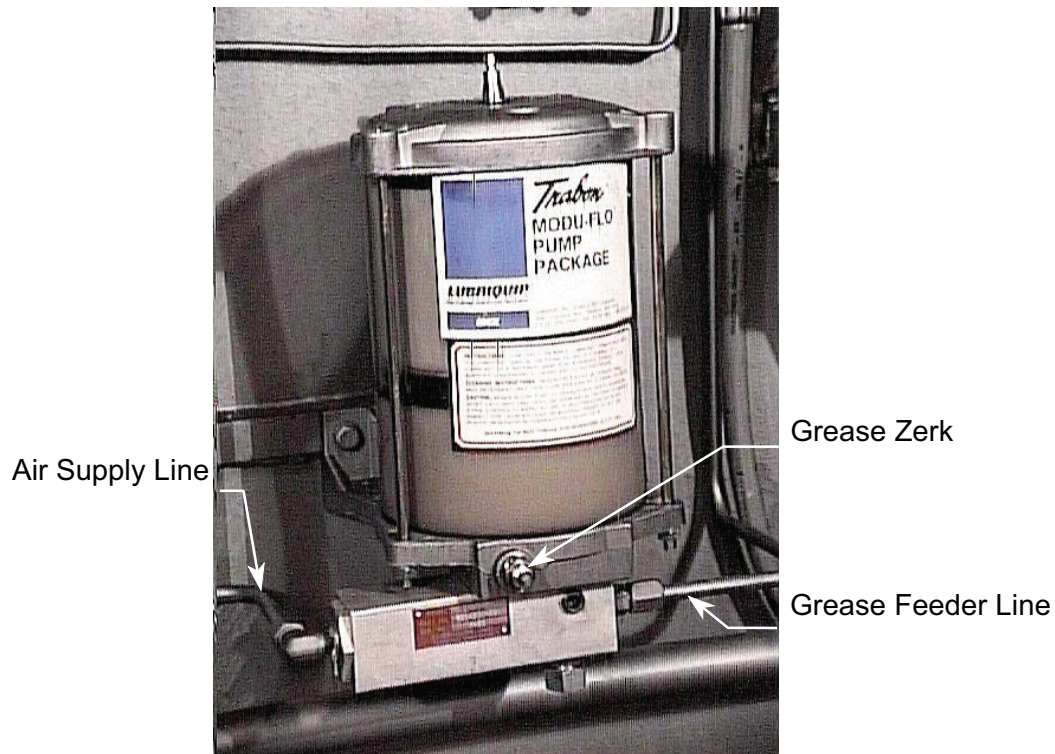


ITEM	QTY	DESCRIPTION
1	2	Lube Hose
2	2	Swivel Fitting
3	1	Pneumatic Pump

- A. Feed Line to RH End
- B. To LH Trunnion Rollers
- C. To LH Agitator Shaft Bearings
- D. Feed Line from LH End
- E. To RH Trunnion Rollers
- F. To RH Gear Case Bearings
- G. To Idler if Chain Drive

SECTION 5-7

PNEUMATIC PUMP OPERATION FIGURE 5-7-2



The **divider valve** is a series progressive unit consisting of modular sections that fill and distribute grease one at a time, in sequence. Positive displacement metering valves in each section of the **divider valve** assure that each section is filled with the correct amount of grease and the grease is sent to its assigned area of lubrication before the next section can be filled.

A **cycle switch** is located on the first section of the **divider valve**. The cycling of grease through the system causes a **cycle pin** to move back and forth activating the **cycle switch**. Once the **cycle pin** moves in or out, all the mixer's lube points must be greased before the pin will move again. An empty grease reservoir, kinked or plugged lines can prevent grease from progressing through the sections of the **divider valve**. Because the **divider valve** cannot complete its cycle, the **cycle pin** remains in its last functioning position.

SECTION 5-7

When **cycle pin** action has been interrupted, a warning is given to the operator that the **Auto Central Lube System** is malfunctioning. The type warning the operator will see depends on the mixer's controls:

- Standard Pushbutton Panel.** Red indicator light labeled **AUTO LUBE** lights up. Indicator light stays lit until the **cycle pin** actuates the **cycle switch**.
- VCP.** Screen displays a warning message for 24 hours while mixer power is on. If mixer is shut off, warning timer stops. When mixer power is restored, warning timer begins again where it left off, and warning message continues. When the 24-hour warning timer times out, the mixer shuts off automatically. Clear the fault or warning by actuating the **cycle switch**, then restart mixer. See **Chapter 2, Options Checklist**, to find out if your mixer has been equipped with a **VCP**.

When the **AUTO LUBE** warning is displayed, the mixer will still run; however, the problem should be solved as soon as possible. If you suspect a blockage in the lube system, see **Section 5-6, Manual Central Lube System**, for instructions on how to locate and correct the problem.

REFILLING THE GREASE RESERVOIR

Check the grease level in the **Auto Central Lube System**'s grease reservoir as part of a weekly maintenance program. To refill, attach a manual or pneumatic grease gun to the zerk or appropriate fitting at the bottom of the grease reservoir. The reservoir is attached to the top of the pneumatic grease pump, which is mounted in the lower right-hand corner inside the mixer's left-hand end door. Pump grease into the bottom of the reservoir, raising the disk follower and spring, until the reservoir is full. See **Figure 5-7-2, Pneumatic Pump Operation**.

AUTO LUBE PUMP PRESSURE RELIEF

At the bottom of the pneumatic pump is a pressure relief port with an orange blowout disc. The disc is designed to blowout if pressure inside the pneumatic pump reaches 2050 PSI. If the system is not getting lubricant, check for grease coming from the bottom of the pneumatic pump. Unscrew the fitting at the bottom of the pump and check the blowout disc inside the fitting. A blown-out disc must be replaced immediately with a new one to restore pressure to the **Auto Central Lube System**. Call the Parts Department at **Peerless Machinery Corp.** for assistance.

MANUAL OPERATION OF THE AUTO CENTRAL LUBE SYSTEM

The **Auto Central Lube System** can be operated manually. To activate the system manually, uncap the zerk fitting mounted on the inboard column inside the left-hand end door. In a typical application, use a standard grease gun to pump approximately 38 strokes (1.5 oz., 2.7 cu. in.) of grease into the zerk. As with the **Manual Central Lube System**, the grease goes into a **divider valve** that distributes it proportionately to each designated area of the mixer.

The pneumatic pump can be cycled manually. Use a screwdriver or other narrow-tipped tool to push a recessed actuator button, marked **AUTO LUBE**, on top of the air solenoid valve. The valve is mounted on the upper-left column inside the left-hand end of the mixer. The pump cycles once for every time the actuator button is depressed.

SECTION 5-8

AGITATOR ASSEMBLY

Peerless supplies mixer agitators according to customer needs. Your mixer has been equipped with a **Double Sigma Arm** arrangement.

See **Figure 5-8-1, Double Arm Agitator Assembly**.

Although the **agitator assembly** requires little maintenance, the following section covers what is required.

SECTION 5-8

DOUBLE ARM AGITATOR ASSEMBLY 5-8-1

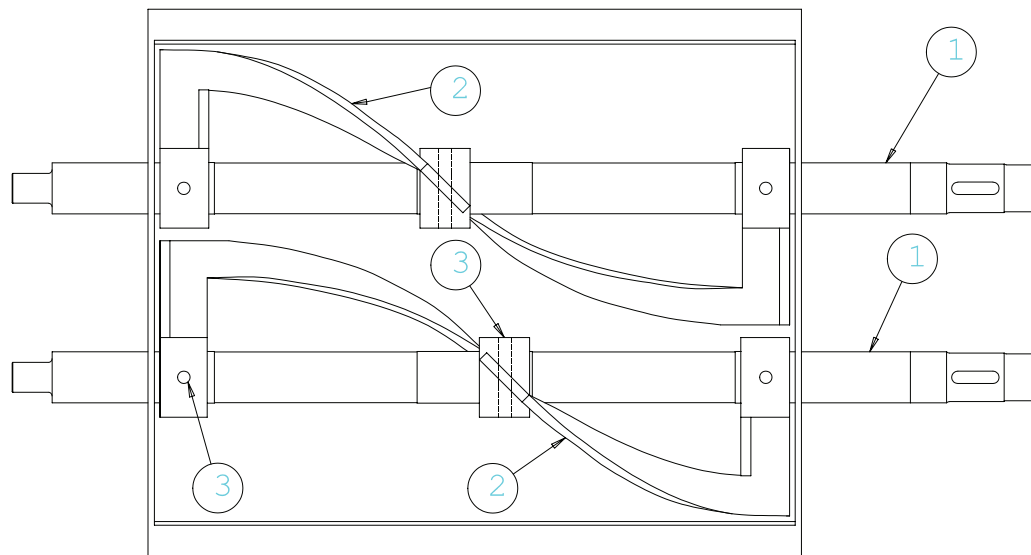
We recommend visual inspection of the following areas once a month:

- ✓ Check agitator-to-bowl clearances.
- ✓ Check agitator for side-to-side motion.
- ✓ Check agitator pin weld for bubbling, cracking.

If problems occur, see **Section 5-9, Left-Hand Agitator Shaft Bearing Assembly**, or call the Service Department at **Peerless Machinery Corp.**

NOTE: Agitator straight pins are stamped with a "P" on each end to show location in hub.

DOUBLE ARM AGITATOR ASSEMBLY FIGURE 5-8-1



ITEM	QTY	DESCRIPTION
1	2	Agitator Shaft
2	2	Mixing Arms
3	6	Straight Pin

SECTION 5-9

LEFT-HAND AGITATOR SHAFT BEARING ASSEMBLY

▲ DANGER

Rotating parts can cause serious injury or death. Disconnect and lockout electrical power before servicing, repairing or otherwise working on mixer. See **Section 1-3, Zero Energy State Procedures**.

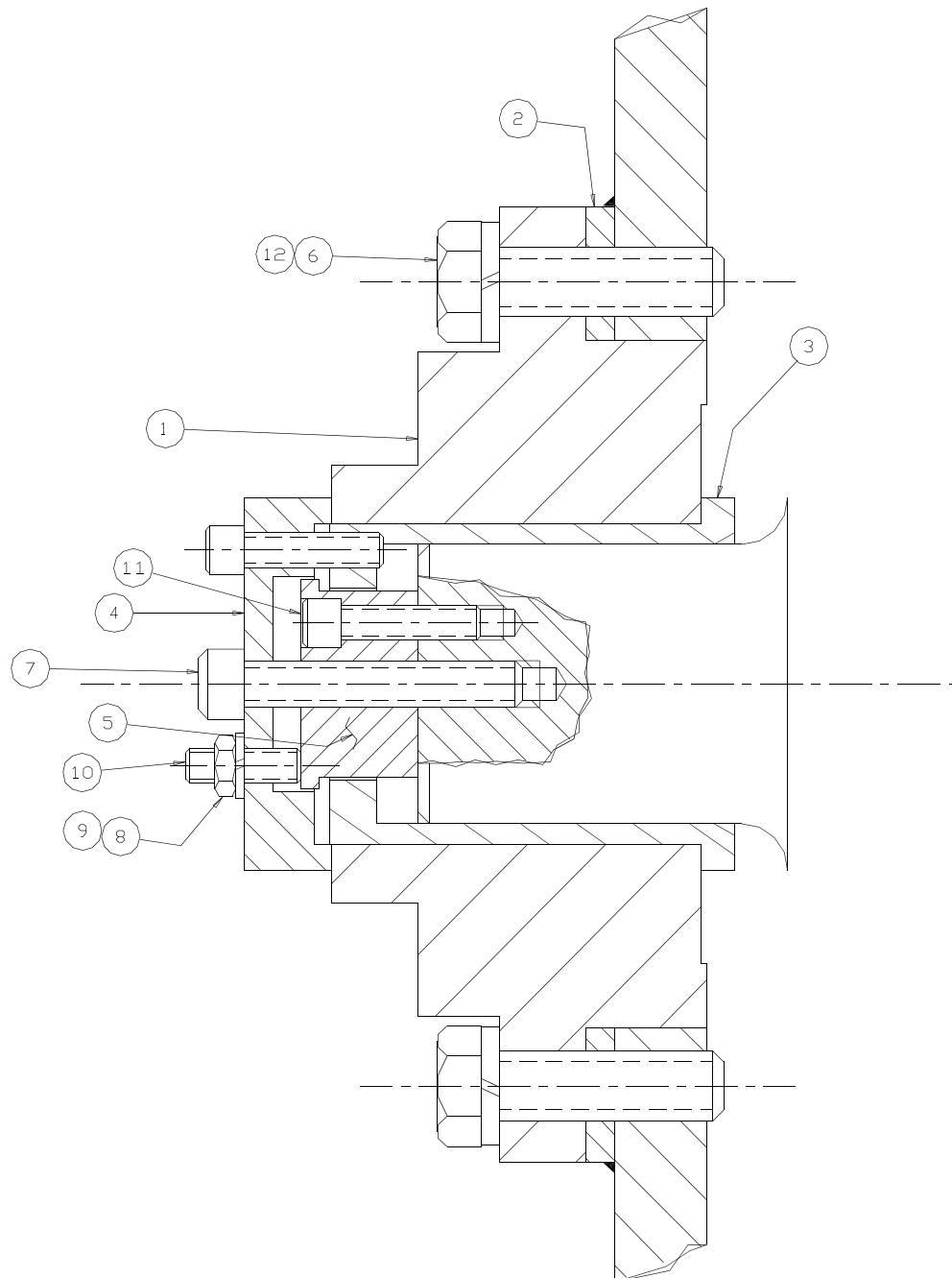
The left-hand agitator shaft bearings assembly consists of roller bearings that support the left-hand end of both agitator shafts. Your mixer is equipped one of two bearing designs, standard or heavy duty, depending on your particular requirements. See **Figure 5-9-1, Left-hand Agitator Shaft Bearing Assembly**, and **Figure 5-9-2, Heavy-Duty Left-hand Agitator Shaft Bearing Assembly**.

At mixer installation and after one week of operation, perform the following preventive maintenance inspections. Thereafter, we recommend making these inspections once a month:

- ✓ Check agitators for side-to-side motion by using a prybar between agitators and bowl end to see if there is movement. Repeat on both sides of agitator. If movement is found, see instructions: **TO CENTER AGITATOR AND SHAFT**.
- ✓ Inspect all seals for damage or wear. Replace as needed.
- ✓ Listen to the **left-hand agitator shaft bearings** running under load for any unusual noise or vibration. Also check for hotter than normal temperatures of bearings and shafts. If there are problems, call the Service Department at **Peerless Machinery Corp**.

SECTION 5-9

LEFT-HAND AGITATOR SHAFT BEARING ASSEMBLY FIGURE 5-9-1



SECTION 5-9

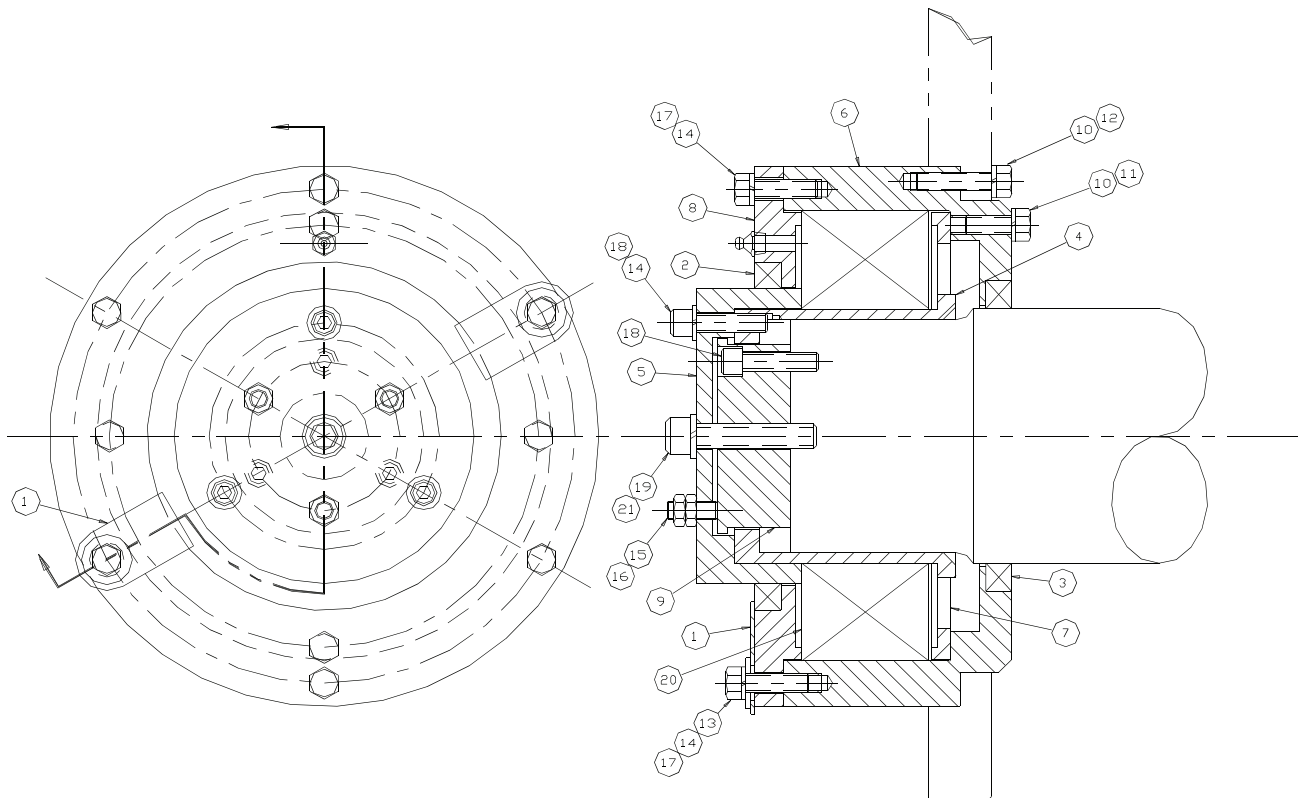
LEFT-HAND AGITATOR SHAFT BEARING ASSEMBLY FIGURE 5-9-1
(con't.)

ITEM	QTY	DESCRIPTION
1	1	Flanged Bearing
2	1	Bearing Spacer
3	1	Inner Sleeve
4	1	End Cap
5	1	Shaft Stop
6	4	Hex Head Cap Screw S/S
7	1	Socket Head Cap Screw M/S
8	3	Lockwasher M/S
9	3	Stop Nut M/S
10	3	Socket Head Set Screw M/S
11	6	Socket Head Cap Screw M/S
12	4	Lockwasher M/S

SECTION 5-9

HEAVY-DUTY LEFT-HAND AGITATOR SHAFT BEARING ASSEMBLY

FIGURE 5-9-2



SECTION 5-9

HEAVY-DUTY LEFT-HAND AGITATOR SHAFT BEARING ASSEMBLY

FIGURE 5-9-2

(con't.)

ITEM	QTY	DESCRIPTION
1	4	Seal Retainer Tab
2	1	JM Split Clipper Seal
3	1	JM Split Clipper Seal
4	1	Inner Sleeve
5	1	End Cap
6	1	LH Shaft Bearing Housing
7	1	Bearing Pushout Ring
8	1	LH Bearing Cover Plate
9	1	Shaft Stop
10	10	Lockwasher S/S
11	4	Hex Head Cap Screw S/S
12	6	Hex Head Cap Screw S/S
13	2	Flat Washer M/S
14	9	Lockwasher M/S
15	6	Hex Jam Nuts M/S
16	3	Socket Head Set Screw M/S
17	6	Hex Head Cap Screw M/S
18	6	Socket Head Cap Screw
19	1	Socket Head Cap Screw
20	11	Spherical Roller Bearing--Double Row
21	1	Lockwasher M/S

TO CORRECT AGITATOR-TO-BOWL INTERFERENCE

If the agitators rub the bowl end, use a prybar to try to move the agitators back and forth. If the agitators do not move, recenter using the procedure: **TO CENTER AGITATOR AND SHAFT**. If you can move the agitators from side to side with a prybar, check to see if the setscrews are loose. If loose, recenter the agitators and lock setscrews down by tightening the nuts. Use the procedure: **TO CENTER AGITATOR AND SHAFT**. See **Figure 5-9-1, Left-Hand Agitator Shaft Bearing Assembly** and **Figure 5-9-2, Heavy-Duty Left-Hand Agitator Shaft Bearing Assembly**.

If the problem still exists, a bearing could be worn. Contact the Service Department at **Peerless Machinery Corp.**

TO CENTER AGITATOR AND SHAFT

See **Figure 5-9-1, Left-hand Agitator Shaft Bearing Assembly** and **Figure 5-9-2, Heavy-Duty Left-Hand Agitator Shaft Bearing Assembly**.

1. Disconnect and lockout power to mixer. See **Section 1-3, Zero Energy State Procedures**.
2.
 - a) To draw the shaft toward you, loosen the three setscrews on the end cap and tighten the socket-head cap screw in the center of the end cap.
 - b) To push the shaft away from you, loosen the socket-head cap screw in the center of the end cap and tighten the three setscrews on the end cap.
3. Check centering and adjust the screws accordingly.
4. After centering the agitator and shaft, make sure all screws are tight. Lock down the jam nuts.
5. Restore power to the mixer.

SECTION 5-10

BOWL MOUNTING ASSEMBLIES

▲ DANGER

Rotating parts can cause serious injury or death. Disconnect and lockout electrical power before servicing, repairing or otherwise working on mixer. See **Section 1-3, Zero Energy State Procedures**.

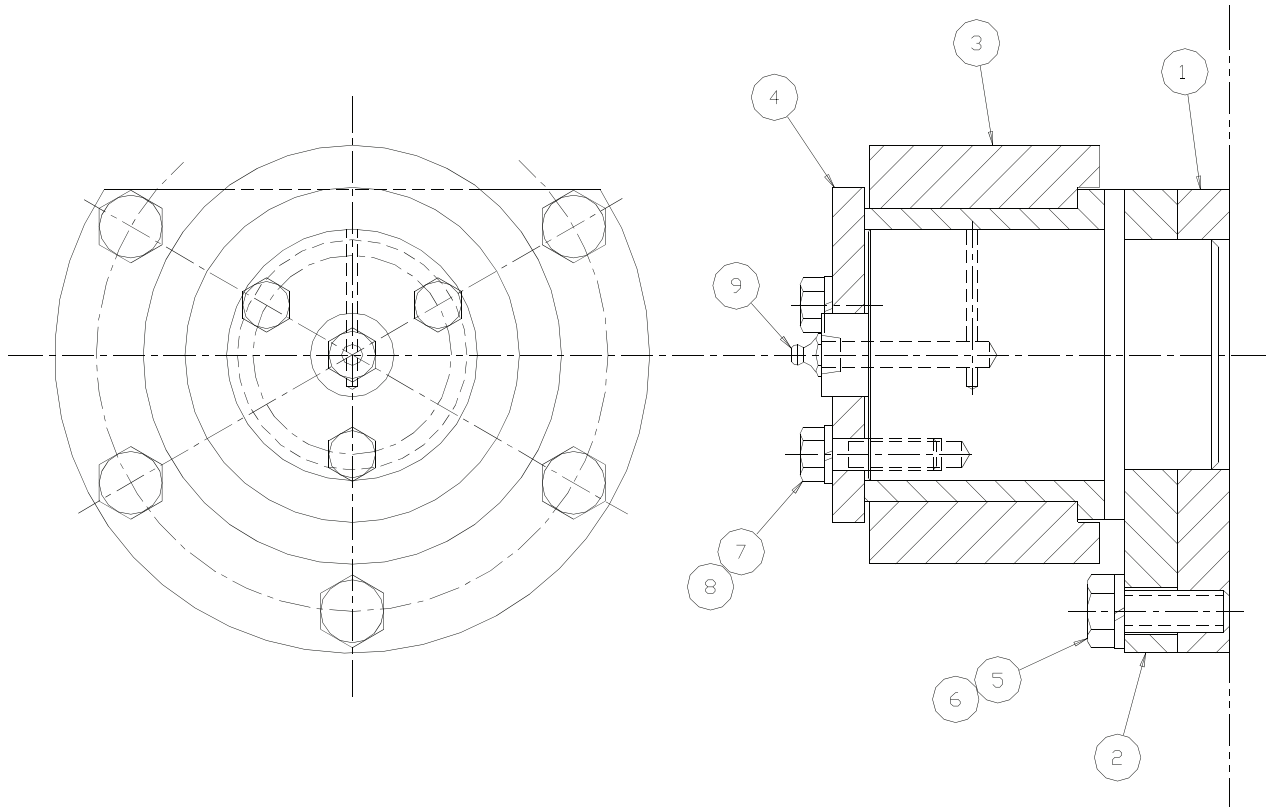
The **bowl mounting assemblies** consist of cam rollers that support the bowl and gear case assemblies. The assemblies consist of a steel roller with a bronze bearing insert riding on a shaft and flange assembly bolted to the mixer frame. See **Figure 5-10-1, Bowl Mounting Assemblies**.

At mixer installation and after one week of operation, perform the following preventive maintenance inspections. Thereafter, we recommend making these inspections once a month.

- ✓ Retorque all bolts. Torque 1/2" bowl mounting bolts to 50 ft-lbs (67.8 Nm). Torque 3/8" bowl mounting bolts to 20 ft-lbs (27.12 Nm).
- ✓ Check for proper lubrication and free rolling.
- ✓ Check for damage and wear. Replace roller if needed.

SECTION 5-10

BOWL MOUNTING ASSEMBLY FIGURE 5-10-1



ITEM	QTY	DESCRIPTION
1	3	Trunnion Roller Bracket
2	1	Trunnion Roller Shaft Assembly
3	1	Trunnion Roller Assembly
4	1	End Plate
5	5	Hex Cap Screw Grade 2
6	5	Lockwasher Grade 5
7	3	Hex Cap Screw M/S
8	3	Lockwasher M/S

SECTION 5-11

BOWL TILT SYSTEM

 **WARNING** Hydraulic oil is flammable and can cause fire. Clean up immediately any hydraulic oil that may have leaked or spilled.

 **WARNING** Hydraulic oil is slippery and can cause falls. Clean up immediately any hydraulic oil that may have leaked or spilled as a result of maintenance.

Your **Peerless** mixer is equipped with a hydraulic system that tilts the bowl to discharge dough.

The standard **Jog and Tilt Feature** incorporated into the tilt system allows you to lower the bowl and jog the agitator at the same time to discharge dough. Using the **Jog and Tilt Feature** results in easier dough discharge and shorter batch cycle times.

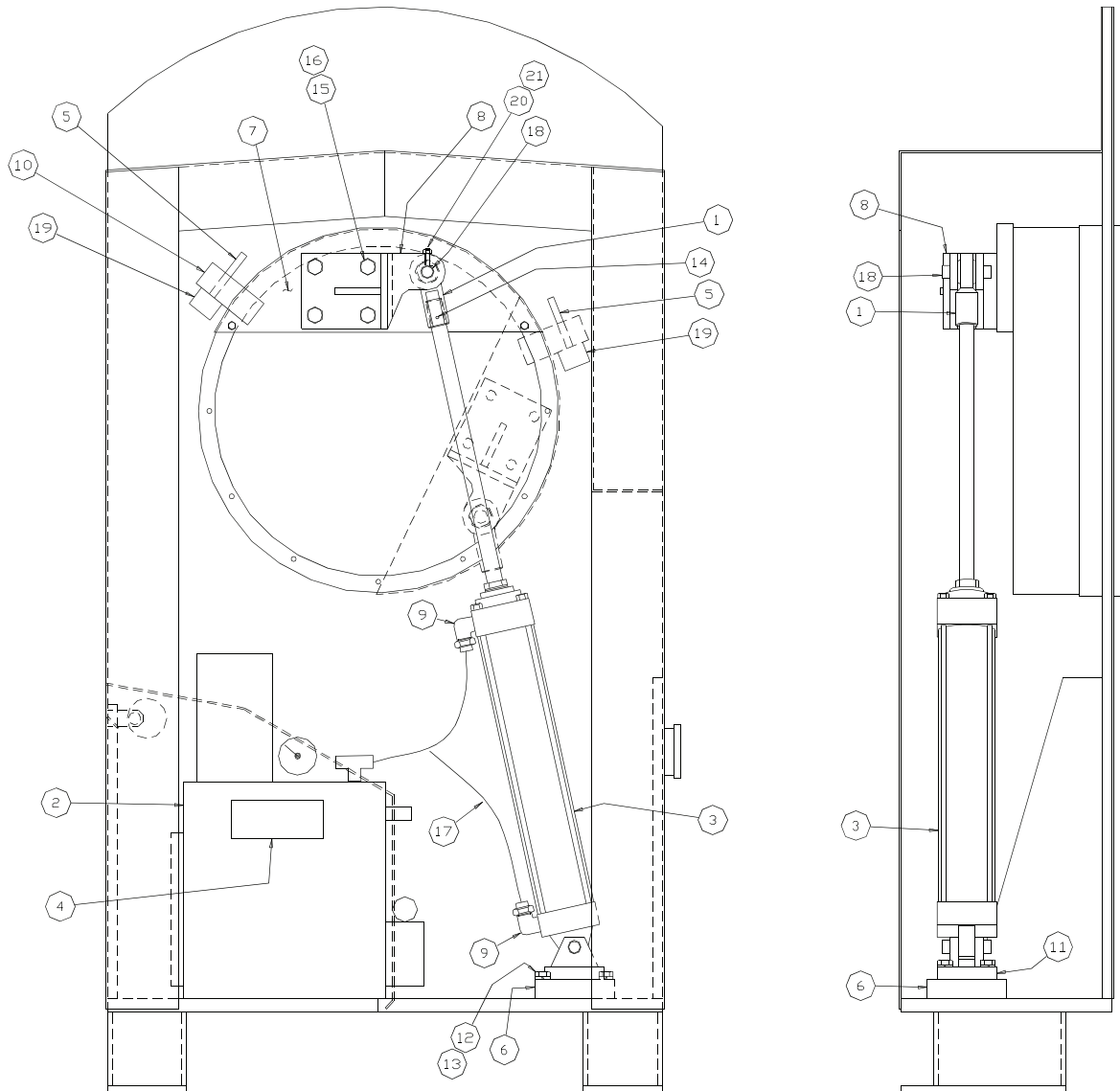
NOTE: When jogging in high speed and tilting bowl down at the same time, the bowl will hesitate at times as it is lowering. This is normal and will not damage mixer.

The hydraulic power unit is inside the left-hand end doors of the mixer. Mixers with 120-degree tilt capabilities have a hydraulic cylinder on the left-hand end. See **Figure 5-11-1**. Mixers with greater than 120-degree tilt have a rotary actuator located inside the left-hand end. See **Figure 5-11-2**.

The following sections describe the main components of the hydraulic power unit. See **Figures 5-11-3, Hydraulic Power Unit** and **5-11-4, Hydraulic Stack**.

SECTION 5-11

HYDRAULIC TILT CYLINDER SYSTEM FIGURE 5-11-1



SECTION 5-11

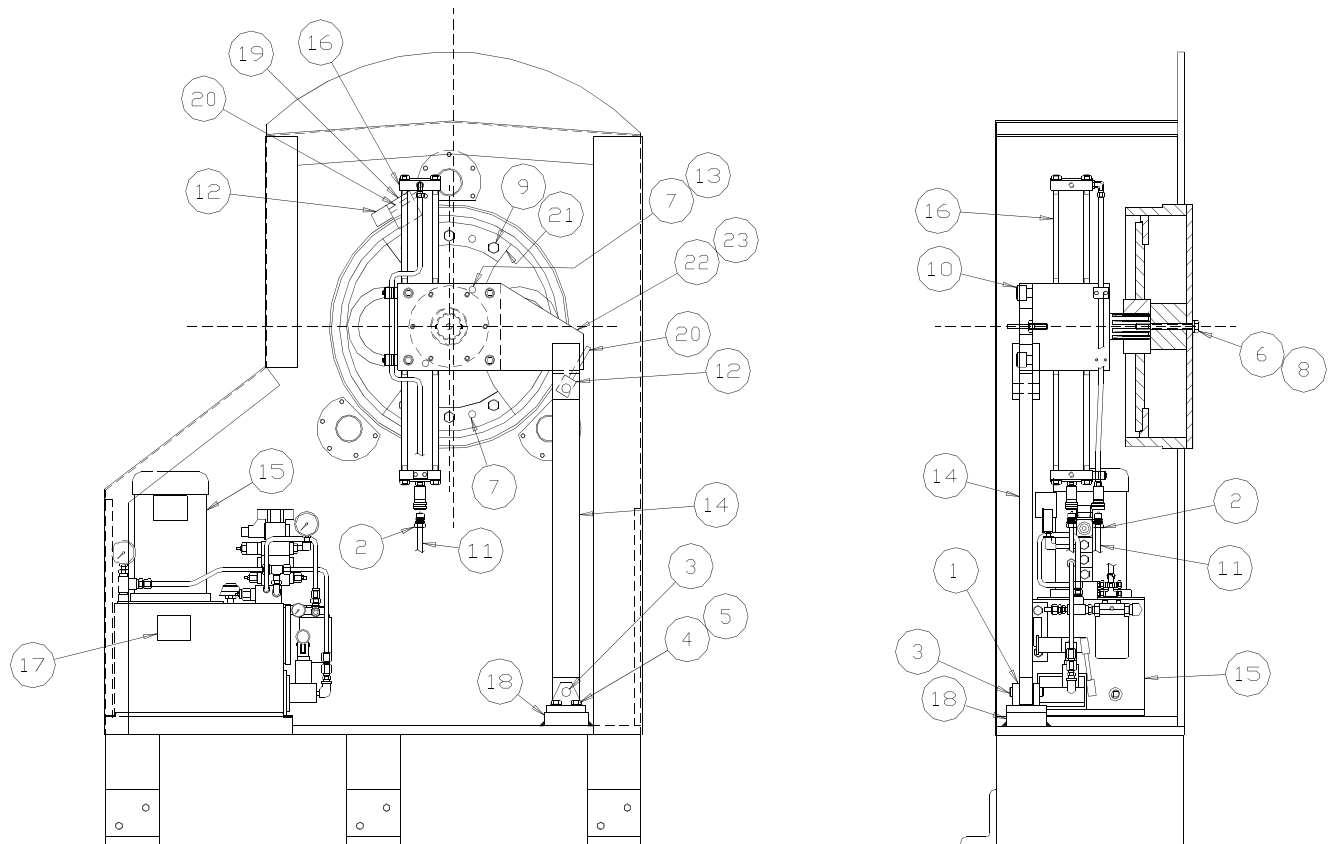
HYDRAULIC TILT CYLINDER SYSTEM FIGURE 5-11-1

(con't.)

ITEM	QTY	DESCRIPTION
1	1	Self-Aligning Rod End
2	1	Hydraulic System
3	1	Hydraulic Cylinder
4	1	Oil Fill Fitting Label
5	1	Gusset
6	1	Cylinder Eye Bracket Mounting Pad
7	1	Tilt Lever Bracket Mounting Pad
8	1	Tilt Lever Bracket Assembly - Machine
9	2	90° Elbow - 1/2 37° JIC x #12 SAE THD
10	1	Bowl Stop Bar
11	1	Eye Bracket - Hydro-Line
12	4	Lockwasher
13	4	Hex Head Cap Screw
14	1	Roll Pin
15	4	Lockwasher
16	4	Hex Head Cap Screw
17	2	Hydraulic Hose Assembly
18	1	Rod End Pin
19	2	Bowl Stop
20	1	Socket Head Set Screw

SECTION 5-11

HYDRAULIC ROTARY ACTUATOR TILT SYSTEM FIGURE 5-11-2



SECTION 5-11

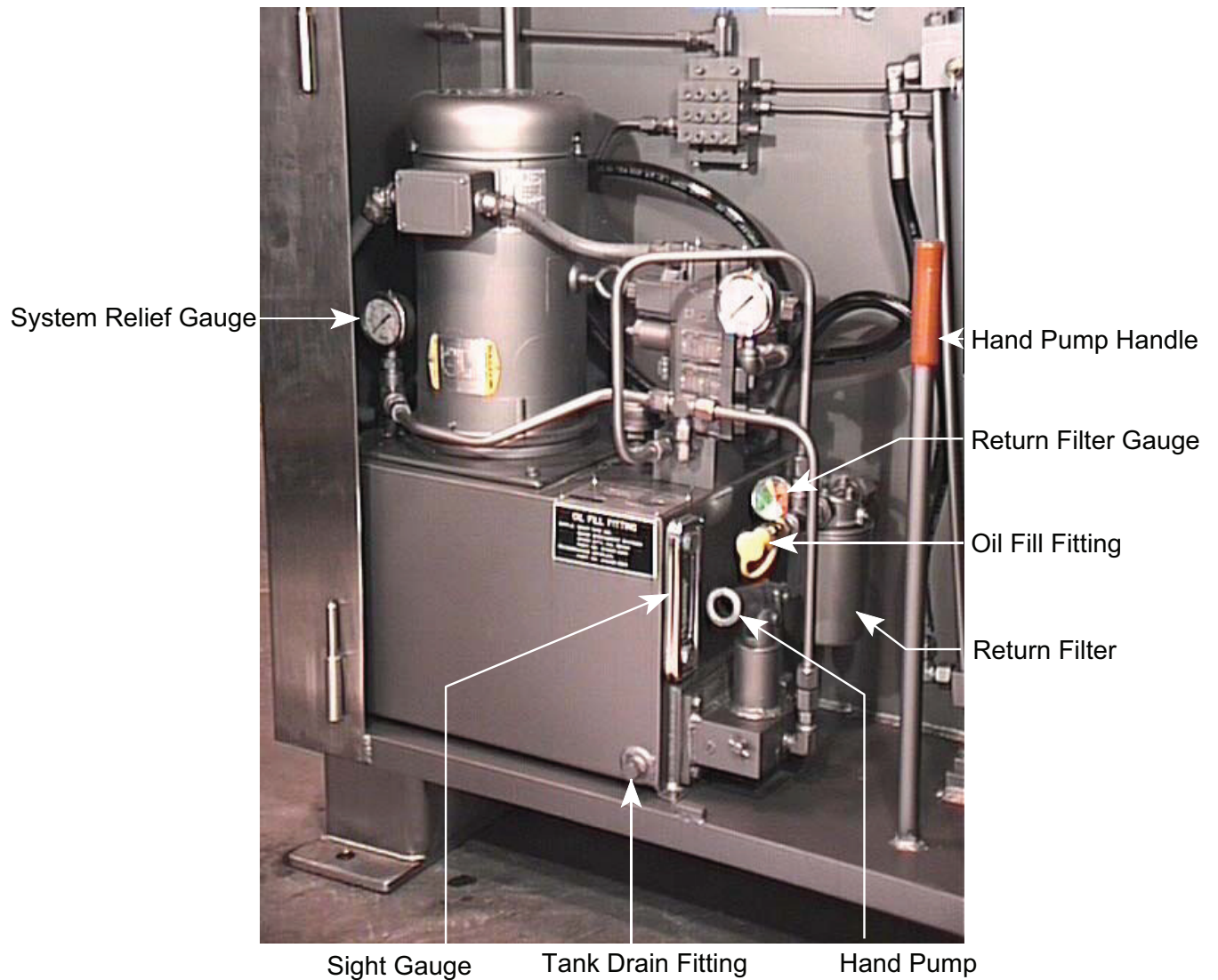
HYDRAULIC ROTARY ACTUATOR TILT SYSTEM FIGURE 5-11-2

(con't.)

ITEM	QTY	DESCRIPTION
1	1	Clevis Bracket
2	2	Male Connector - Weatherhead - 37 deg JIC x 3/8 NPT
3	2	Pivot Pin
4	4	Lockwasher
5	4	Socket Head Cap Screw
6	7	Lockwasher
7	6	Dowel Pin
8	1	Hex Head Cap Screw
9	6	Hex Head Cap Screw
10	4	Socket Head Cap Screw
11	2	Hydraulic Hose Ass'y - 1/2 Determine length at ass'y
12	2	Bowl Stop Bar
13	1	Actuator and Arm Pinning
14	1	Torque Support Bar
15	1	Hydraulic System
16	1	Rotary Actuator - FLO TORK
17	1	Oil Fill Fitting Label
18	1	Clevis Bracket Mounting Pad
19	1	Bowl Stop Block
20	2	Gusset
21	1	Tilt Lever Arm - PMC Machine
22	1	Actuator Lever Arm - PMC Machine
23	1	Lever Arm - Weld

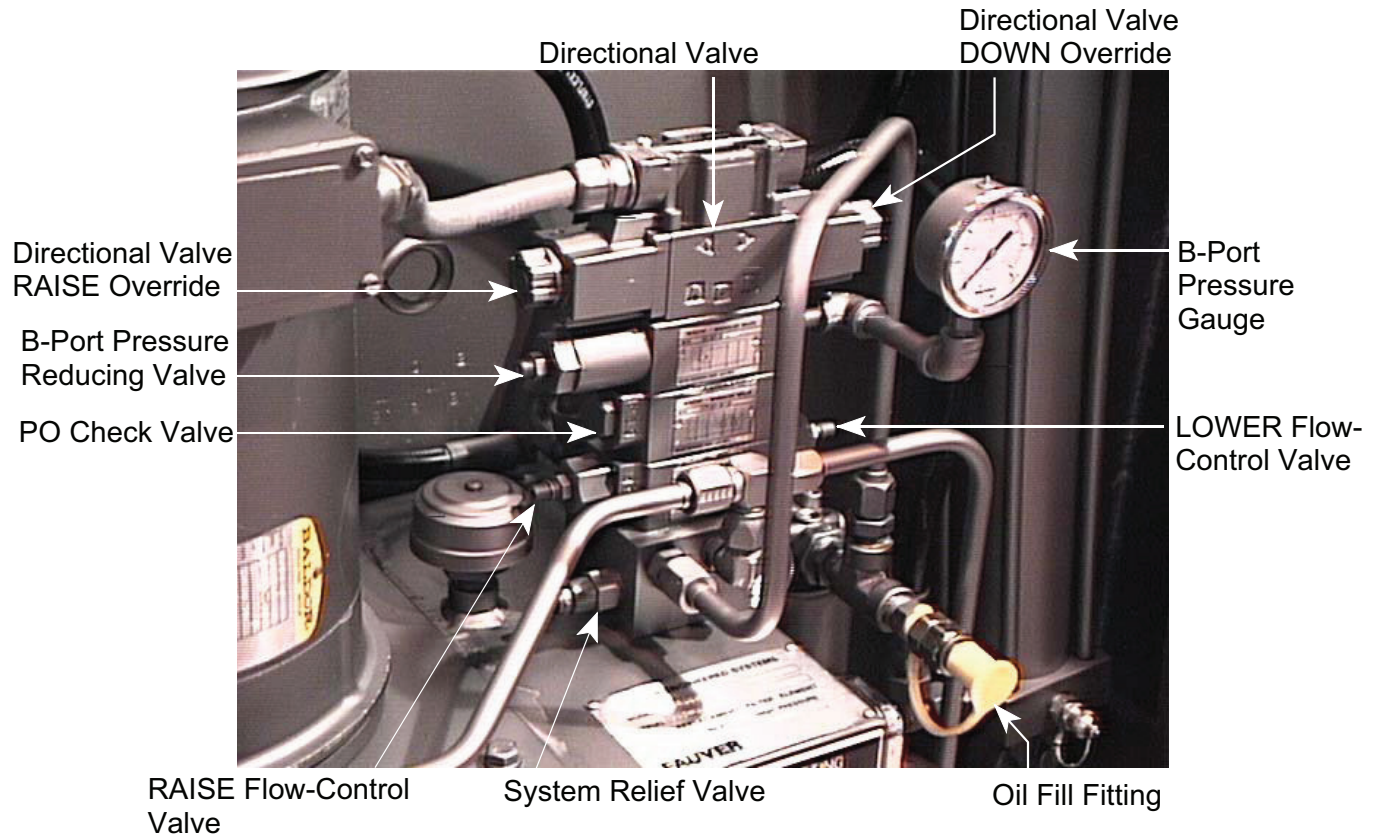
SECTION 5-11

HYDRAULIC POWER UNIT FIGURE 5-11-3



SECTION 5-11

HYDRAULIC STACK FIGURE 5-11-4



SECTION 5-11**SYSTEM RELIEF VALVE 5-11-1**

The **system relief valve** controls the system's hydraulic oil pressure. For 120-degree tilt systems, pressure is set at 1500 PSI. If your mixer has greater than 120-degree tilt, see **Table 5-11-5, Hydraulic System Pressure Settings**, for the pressure setting specific to your mixer.

TO ADJUST HYDRAULIC PRESSURE SETTING

To adjust the hydraulic pressure, use the pressure relief valve and the flow control valves above the hydraulic fluid reservoir, or tank.

1. Screw in the **lower** flow control valve until bowl stops moving.
2. System pressure gauge shows the relief valve pressure setting.
3. Screw *in* the relief valve screw to increase pressure. Screw *out* the relief valve screw to decrease pressure.
4. Once pressure is set, lock down with the lock nut and readjust the flow control valve.

SECTION 5-11

HYDRAULIC SYSTEM PRESSURE SETTINGS TABLE 5-11-5

FOR ROTARY ACTUATOR SYSTEM ONLY

DA MODEL NUMBER (40180 RPM)	SYSTEM PRESSURE RELIEF SETTING
DA100 DA100HD DA100FD	2100 PSI 2100 PSI 2100 PSI
DA150 DA150HD DA150FD	2100 PSI 2100 PSI 2100 PSI
DA200 DA200HD DA200FD	2100 PSI 2100 PSI 2100 PSI
DA250 DA250HD DA250FD	2100 PSI 2100 PSI 2100 PSI
DA300 DA300HD DA300FD	1800 PSI 1800 PSI 1800 PSI
DA350 DA350HD DA350FD	1800 PSI 1800 PSI 2700 PSI
DA400 DA400HD DA400FD	2200 PSI 2200 PSI 2200 PSI
DA500 DA500HD DA500FD	2200 PSI 2200 PSI 2200 PSI
DA600 DA600HD DA600FD	2200 PSI 2200 PSI 2200 PSI
DA650 DA650HD DA650FD	2200 PSI 2200 PSI 2200 PSI

SECTION 5-11

FLOW-CONTROL VALVES 5-11-2

The **flow-control valves** control back-pressure of the hydraulic fluid in the system. The **flow-control valves** restrict the flow of hydraulic fluid as it returns from the cylinder or rotary actuator to the tank. The restriction increases the pressure of the return fluid, which increases the force moving the cylinder or rotary actuator. This increased operating force may be required to smooth out bowl movement as it is being tilted.

The **flow-control valves** will also control the speed of the bowl movement.

The hydraulic system has separate **flow-control valves** for bowl-up and bowl-down movement. The valves are usually set *full open*, or, with the adjusting screws screwed *out*.

Adjustment may be necessary to obtain smooth bowl movement. If the bowl jerks when being lowered with no product inside, follow these steps to adjust the *down* flow control:

1. Screw *in* adjusting screw until bowl just starts to slow down. At this setting, the bowl should lower with no, or very little, jerking.
2. Lock flow-control adjusting screw using lock nut.

PRESSURE OPERATED CHECK VALVES (PO CHECKS) 5-11-3

The **pressure operated check valves (PO checks)** trap hydraulic fluid in the cylinder or actuator holding the bowl in the position it was stopped at. No adjustments are necessary.

B-PORT PRESSURE REDUCING VALVE (120-degree systems only) 5-11-4

On 120-degree systems, the **B-port pressure-reducing valve** controls the pressure in the **B-port**. The **B-port** supplies the pressure to extend the cylinder and raise the bowl. The **B-port** is always set at 800 PSI to prevent damage to the cylinder while extending.

To Adjust B-Port Pressure:

1. Screw *in* the UP flow-control valve until the bowl stops moving.
2. The **B-port** pressure gauge shows the current setting of the **B-port pressure-reducing valve**.
3. Screw *in* the adjusting screw on the **B-port pressure-reducing valve** to *increase* the pressure. Screw *out* the adjusting screw on **B-port pressure-reducing valve** to *decrease* the pressure.
4. When the pressure gauge reaches 800 PSI, lock down the adjusting screw using the lock nut.
5. Readjust the UP flow-control valve.

DIRECTIONAL VALVE 5-11-5

The **directional valve** directs the flow of hydraulic fluid to the cylinder or rotary actuator to raise or lower the bowl. The **directional valve** is operated by solenoid valves. When the solenoid valves are energized, one of the indicator lights will light up depending on which directional valve has been activated.

You can manually actuate the **directional valves**. Use the round handle on top of the solenoid valve of the **directional valve**. See **Section 5-11-14, Manual Tilt Procedure**. *Do not manually override if the solenoid is energized.*

RETURN FILTER 5-11-6

The **return filter** filters the hydraulic oil returning to the tank. A gauge in front of the **return filter** shows when the filter should be changed. If the needle goes into the yellow range while the bowl is moving, change the **return filter**. *The gauge works only when the bowl is moving.*

To Change the Hydraulic Filter:

1. Disconnect and lock out power to the mixer. See **Section 1-3, Zero Energy State Procedures**.
2. Unscrew filter from mounting flange.
3. Screw on new filter.

NOTE: Peerless supplies a spare filter with each new mixer. The spare filter is shipped loose in a plastic bag, attached to the hydraulic unit.

OIL FILL FITTING 5-11-7

The **oil fill fitting** is located in front of the oil filter and is used to fill the hydraulic system with hydraulic fluid. See **Table 5-2-5, Recommended Lubricants Chart**, for the type of hydraulic fluid recommended for your mixer.

We strongly recommend you fill the tank through the **oil fill fitting** so the oil goes through the filter before going into the tank. The oil from most suppliers is not clean enough to use on a hydraulic system without being filtered. In general, the biggest cause of hydraulic system failure is dirty oil. Always use clean hydraulic fluid.

To Add Hydraulic Fluid:

1. Connect the fill coupling provided to an electric or hand pump and hose that are supplied by the user.

NOTE: The fill coupling included is shipped loose with the spare hydraulic filter in a plastic bag, attached to the hydraulic unit. The coupling is a Faster part, number TNV06-NPT-FS.

2. Install the suction line of the user-supplied pump into a clean supply of hydraulic fluid.

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3. Connect the fill coupling to the **oil fill fitting**.
4. Fill the tank until the hydraulic fluid level is 2" from the top of the tank. Use the sight gauge to monitor the oil level.
5. Cycle the bowl up and down, then add more fluid if needed.

SIGHT GAUGE 5-11-8

The sight gauge on the right-hand side of the tank shows the level of hydraulic fluid in the tank. The proper level for the fluid is approximately 2" from the top of the sight gauge. However, because of the displacement of the rods on the tilt cylinder, the fluid level will vary slightly depending on the position of the bowl.

TANK DRAIN FITTING 5-11-9

The tank drain fitting is located at the front of the hydraulic fluid tank next to the edge of the base plate. Unscrew the drain fitting to drain hydraulic fluid from the tank.

Drain the hydraulic fluid, flush the tank, and add new oil every six months.

HAND PUMP 5-11-10

The **hand pump** is used to raise or lower the bowl manually. For storage, the pump handle slides onto a stud in the front, right-hand corner of the mixer base plate.

Use the **hand pump** to raise or lower the bowl in an emergency. See **Section 5-11-14, Manual Tilt Procedure** , for steps to tilt the bowl using the **hand pump**.

SECTION 5-11**TILT CYLINDER 5-11-11**

A hydraulic **tilt cylinder** on the left-hand end of the mixer is used to tilt the bowl on mixers with 120-degree tilt capability.

Cylinder rod ends are cushioned to slow the tilt speed as the bowl approaches the full-up position. Slowing the tilt speed allows the bowl to stop smoothly and consistently in exactly the same position. Manually grease the rod ends once a month.

ROTARY ACTUATOR 5-11-12

A **rotary actuator** raises and lowers the bowl, instead of a tilt cylinder, on mixers with greater than 120-degree tilt capability (Over Tilt). See **Chapter 2, Options Checklist** to find out if your mixer has Over Tilt. The actuator is located inside the left-hand end door of the mixer.

A **relief valve** is mounted on the top edge of the center housing. If oil flows from the **relief valve**, the piston seals are leaking and need to be replaced. Call the Service Department at **Peerless Machinery Corp.** for assistance.

TILT LIMIT SWITCHES 5-11-13

Two **tilt limit switches**, mounted on the inboard column to the left of the cover plate, are used on standard mixers to limit bowl travel. The upper **limit switch** shuts off the hydraulic pump and directional valve coil when the bowl reaches its full-up position. The lower **limit switch** shuts off the hydraulic pump and directional valve coil when the bowl reaches its full-down position.

Peerless supplies other tilt switches at customer request for various functions. See **Chapter 2, Options Checklist**, for other tilt switches that may be included with your mixer.

To Adjust Tilt Limit Switches:

If the bowl hits the UP or DOWN stops too hard or does not hit them at all, use the following procedure to adjust the **tilt limit switches**:

1. Run the bowl down until the tilt lever arm is almost to the bowl stop.
2. Slide the target bracket in front of the lower **limit switch**. If the indicator light on the **limit switch** does not light, loosen the nuts on the **limit switch** and slide it toward the target until the indicator lights up. Retighten the nuts with the **limit switch** in this position.

NOTE: If indicator does not light when target is about 1/4" from face of limit switch, the switch may be defective or disconnected.

3. Loosen the screws on the target. Slide the target counterclockwise until the light on the **limit switch** goes out.
4. Slide the target slowly clockwise until the indicator light just comes on, and tighten in this position.
5. Raise the bowl and relower to check adjustment. You may have to reposition the target to maintain a slight space between the tilt lever arm and the bowl stop.
6. Raise the bowl until it almost seals against the canopy.
7. For the upper **limit switch**, adjust the **limit switch** until the indicator lights as in step 2.
8. Slide the target bracket clockwise until the indicator on the **limit switch** goes out, then slide counterclockwise until the indicator just comes on. Tighten in this position.
9. Readjust as necessary to obtain the proper seal between bowl and canopy, and so the bowl closes without undue force.

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MANUAL TILT PROCEDURE 5-11-14

The bowl can be tilted manually if desired with the **hand pump** and handle provided. See **Section 5-11-10, Hand Pump** for details about the **hand pump**.

1. Use the end of the pump handle to close the needle valve at the bottom, front of the hand pump.
2. Insert the handle into the hand pump.
3. Manually override the directional valve to move the bowl using a narrow-tipped tool (screwdriver, etc.). *Press and hold the small pin in the center of the solenoid coil all the way in.*
4.
 - a) Lower the bowl by pressing and holding in the pin on the solenoid coil labeled (right side), while pumping up and down with the pump handle.
 - b) Raise the bowl by pressing and holding in the pin on the solenoid coil labeled (left side), while pumping up and down with the pump handle.



TILT SYSTEM TROUBLE-SHOOTING GUIDE 5-11-15

Should problems occur with the tilt system, start with **Table 5-4-1, Maintenance Trouble-Shooting Guide** to find and solve the problem. If you need more help, refer to the table below. If the problem cannot be solved or is not listed in the table below, call the Service Department at **Peerless Machinery Corp.**

NOTE: For some trouble-shooting problems, it may be necessary to use the pressure testing ports on the cylinders. **Peerless** has gauges with fittings to connect to the ports that we will ship to our customers for testing. To request the gauges, call the Service Department at **Peerless Machinery Corp.**

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TILT SYSTEM TROUBLE-SHOOTING GUIDE TABLE 5-11-6

TROUBLE	CAUSE	REMEDY
Bowl jerks when tilting without dough.	Too little system pressure.	Adjust system relief pressure per Section 5-11-1 .
	Too little back pressure.	Adjust flow control valves per Section 5-11-2 .
	Cam rollers need greased.	Check Central Lube System, Section 5-6 or 5-7 .
Bowl travels very slowly.	Flow controls adjusted too slow.	Readjust flow control valves per Section 5-11-2 .
	Pump worn out.	Replace.
	Piston seals in cylinder leaking.	Replace.
	Return filter plugged.	Replace.
Bowl does not stay in mid position (Settles).	PO Check valve dirty or bad.	Clean or replace.
	Piston seals in cylinder leaks.	Replace.
	Leak in hydraulic line or hose.	Locate and fix.
Chatter noise coming from cylinder.	Bowl moving too slowly.	Readjust flow control valve per Section 5-11-2 .
	Not enough system backpressure.	Readjust flow control valve per Section 5-11-2 .
	Bad cylinder.	Replace.
	Seals in cylinder worn.	Replace.
No system pressure.	Pump bad.	Replace.
	Pump to motor coupling bad.	Repair or replace.
	Directional valve coil not energized.	Repair.
	Directional valve coil bad.	Replace.
	Low hydraulic fluid.	Add.

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DRIVE SYSTEM

The **drive system** consists of the **drive chain** or optional **drive belt** and sprockets, **gear reducer**, **motor coupling**, **drive motor** and **gear case assembly**. See **Figure 5-12-1, Chain Drive System**, or **Figure 5-12-5, Belt Drive System**. In a typical application, components of the **drive system** on a **Peerless** mixer are located inside the right-hand end doors and under the drip pan.

 DANGER

Before servicing, repairing, or cleaning mixer, follow **LockOut/TagOut** procedure as required by **OSHA Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

See **Section 5-1, Start-Up Checklist**, for recommended areas to be checked at mixer start-up and after one week of operation. See also **Section 5-2, Preventive Maintenance**, for the recommended preventive maintenance schedule for **drive system** components.

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CHAIN DRIVE SYSTEM 5-12-1

The **drive chain** is a multi-strand roller chain, and is sized according to **drive motor** HP and final agitator speed. The **drive chain** further reduces drive motor speed from the **gear reducer** to the **agitator shaft**. See **Figure 5-12-1, Chain Drive System**.

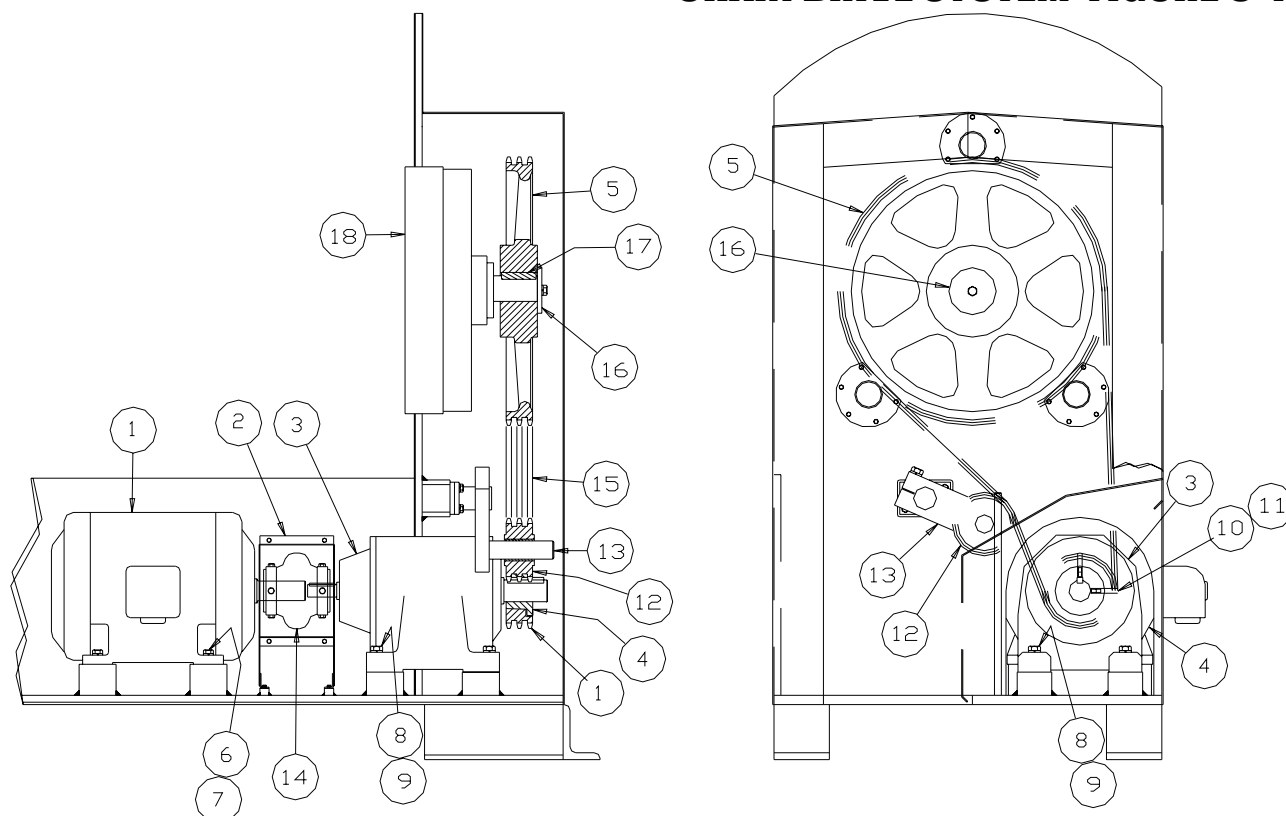
The **drive chain** is lubricated automatically by an **electric oiler** which is mounted on the inboard column above the **drive sprocket**. See **Figure 5-12-2, Electric Chain Oiler**. An electric **solenoid valve** opens allowing oil to flow from the reservoir onto the **drive chain**. Oil flows only when the mixer is running in low speed to prevent excess oil sling and reduces waste. A **flow control valve** on the outlet of the oiler controls the rate of oil flowing to the chain. The valve is factory preset at 10 drops per minute. The flow rate may need adjusted depending on how long the mixer runs in low speed. Turn the **flow control valve** in or clockwise to reduce the flow. Adjust so the **drive chain** is coated with fresh oil and there is little fresh oil on the base plate under the **drive chain**.

As part of a weekly maintenance schedule, check the one-pint capacity reservoir on top of the oiler. Fill with oil as needed through the fill opening on top of the oiler. See **Table 5-2-5, Recommended Lubricants Chart** for type of oil to use.

The **drive chain** is preset with an initial slack equal to the approximate thickness of the chain, but the chain can be expected to stretch after the first week of operation. Tighten the chain if slack exceeds twice the thickness of the chain. See **Figure 5-12-3, Roller Chain Tension**.

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CHAIN DRIVE SYSTEM FIGURE 5-12-1



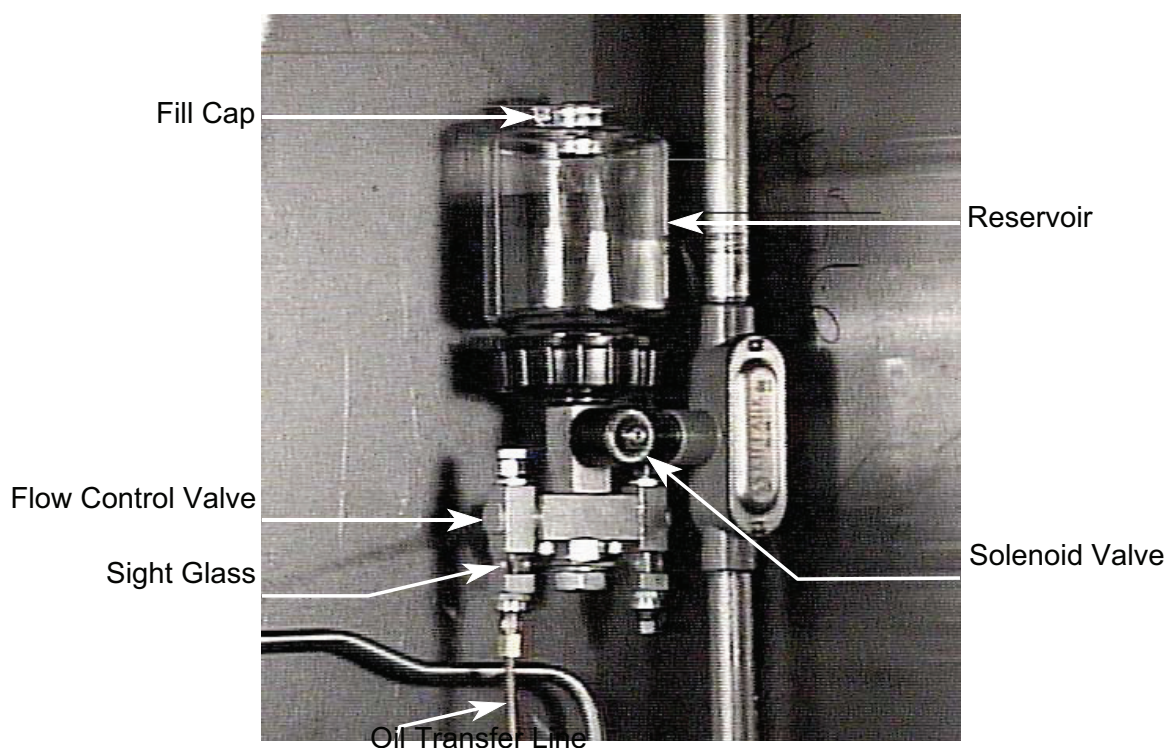
ITEM	QTY	DESCRIPTION
1	1	Drive Motor
2	1	Coupling Guard
3	1	Reducer
4	1	Reducer Sprocket
5	1	Agitator Sprocket
6	4	Hex Head Cap Screw--Motor
7	4	Lockwasher M/S--Motor
8	4	Hex Head Cap Screw--Reducer
9	4	Lockwasher M/S--Reducer
10	4	Socket Head Set Screw
11	4	Socket Head Hollow-Lock Set Screw
12	1	Idler Sprocket
13	1	Idler Arm Assembly
14	1	Motor Coupling

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CHAIN DRIVE SYSTEM FIGURE 5-12-1 (con't.)

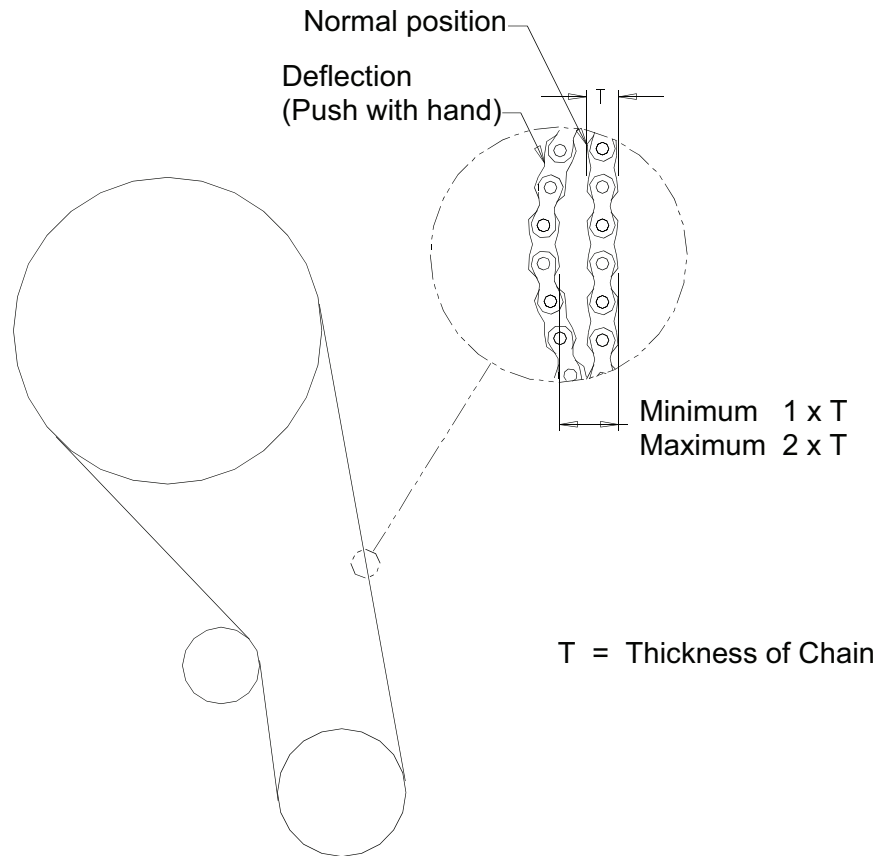
ITEM	QTY	DESCRIPTION
15	1	Drive Chain
16	1	Sprocket End Cap
17	1	Key
18	1	Gearcase Assembly

AUTOMATIC ELECTRIC CHAIN OILER FIGURE 5-12-2



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ROLLER CHAIN TENSION FIGURE 5-12-3



TO TIGHTEN DRIVE CHAIN:

Loosen hex-head bolt on idler arm. Move sprocket into drive chain. Tighten bolt.

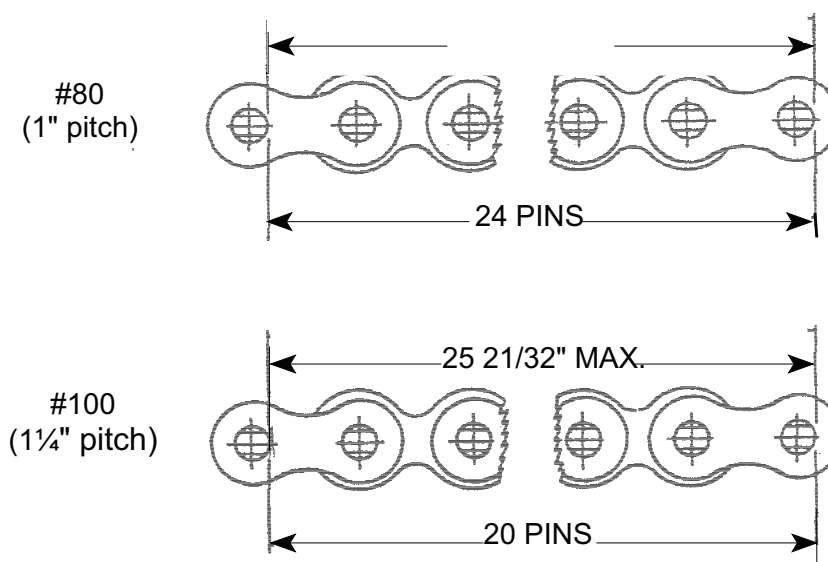
NOTE: Be careful not to adjust drive chains too tight. Tight-running chains tend to become noisy and will decrease life of chain and sprockets.

TO REPLACE DRIVE CHAIN:

During operation, the **drive chain** stretches and wears over time. To reduce sprocket wear, replace the **drive chain** if expansion is more than 3 percent of the nominal. See **Figure 5-12-4, Roller Chain Wear**.

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ROLLER CHAIN WEAR FIGURE 5-12-4



DRIVE BELT SYSTEM (OPTIONAL) 5-12-2

Peerless can equip your mixer with an optional **drive belt system**. See **Chapter 2, Options Checklist**, to find out if your mixer has a drive belt.

The body and teeth of the **drive belt** are made of polyurethane specially designed to adhere to rows of woven cords that run through its length. The drive sprocket has flanged outer edges which guide the belt along its track and over the unflanged agitator sprocket. See **Figure 5-12-5, Drive Belt System**.

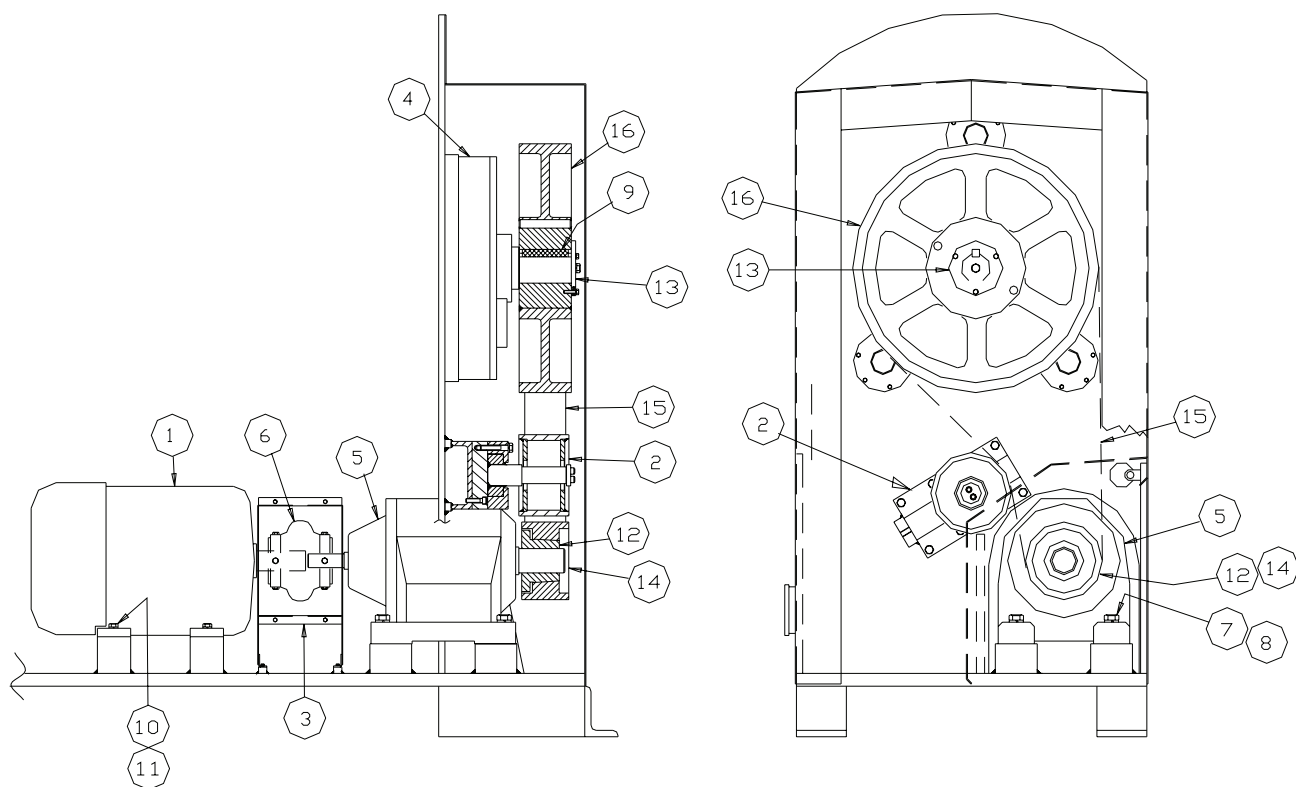
Before shipment, **Peerless** tensions the **drive belt** on each mixer using a sonic tensioner device. See **Chapter 2, Options Checklist**, for the tension value specific to your mixer's **drive belt**.

Since these belts virtually do not stretch, the **drive belt** on your mixer may never need retensioning. However, we strongly recommend that on mixer installation, you handle the belt to get a feel for its tension. When you push hard on a correctly-tensioned belt with your hand, the belt should deflect no more than 1/4" from its original position.

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NOTE: To reduce chances of cracking, do not backbend, crimp, or twist belts when handling. Do not coil belts using a smaller than 5" radius.

BELT DRIVE SYSTEM FIGURE 5-12-5



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BELT DRIVE SYSTEM FIGURE 5-12-5

(con't.)

ITEM	QTY	DESCRIPTION
1	1	Drive Motor
2	1	Idler Assembly
3	1	Coupling Guard
4		Gearcase Assembly
5	1	Parallel Reducer
6	4	Rex Omega Coupling
7	4	Lockwasher--Reducer
8	1	Hex Head Cap Screw--Reducer
9	1	Key
10	4	Lockwasher--Motor
11	4	Hex Head Cap Screw--Motor
12	1	Taperlock Bushing
13	4	Sprocket End Cap
14	1	Reducer Pulley
15	1	Drive Belt
16	1	Agitator Pulley

TO ADJUST DRIVE BELT TENSION:

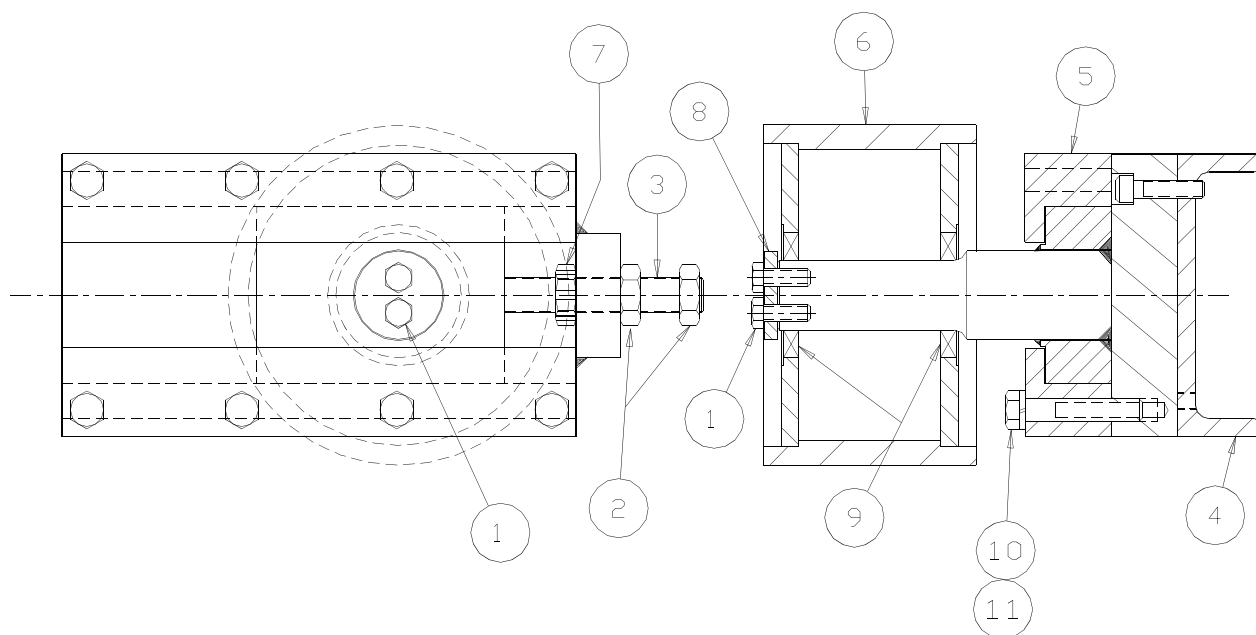
If it becomes necessary to tension the belt, follow the steps below. See **Figure 5-12-6 Belt Idler Operation**.

1. Loosen all eight bolts on the idler mounting bracket. See **Items 10** and **11**.
2. Loosen the jam nut that holds the adjusting rod in place. See **Items 2** and **7**.
3. Tighten the adjusting rod to push the idler toward the belt.
4. Retighten the idler mounting bracket bolts. Torque to 150 ft/lbs.

NOTE: The Gates Rubber Company is working to develop a hand-held mechanical tensioner device. When the new device is available, **Peerless** will make it available to you, our customer.

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BELT IDLER OPERATION FIGURE 5-12-6



ITEM	QTY	DESCRIPTION
1	2	Socket Head Cap Screw
2	2	Hex Nut
3	1	Idler Adjusting Screw
4	1	Bracket Mounting Pad
5	1	Idler Bracket Assembly
6	1	Belt Idler Assembly
7	1	Slotted Hex Nut
8	1	Idler Mounting Cap
9	2	Conrad Bearings
10	8	Lockwasher
11	8	Hex Head Cap Screw

SECTION 5-12

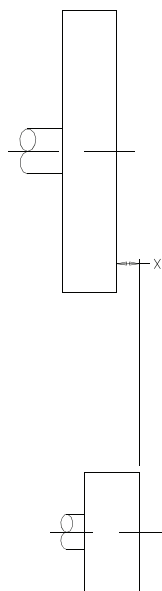
TO CHECK AND ADJUST DRIVE BELT ALIGNMENT:

Peerless aligns each **drive belt** system before shipping the mixer. Under normal mixer operation, the **drive belt** should not need realigned.

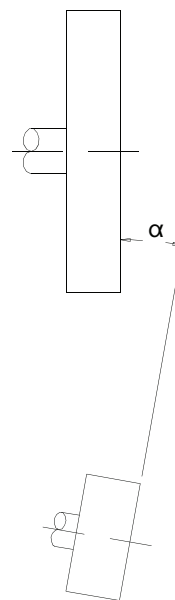
NOTE: During normal operation, **drive belts** tend to track toward the inner or outer edge of the drive sprocket. A belt riding off-center and lightly rubbing the flange is common and should not cause alarm.

To operate correctly, the **drive belt** on your mixer must be in both linear and angular alignment. Angular alignment means the sprocket centers are parallel with each other. Linear alignment means the agitator sprocket and drive sprocket are even with each other. See **Figure 5-12-7, Belt Drive Misalignments**.

BELT DRIVE MISALIGNMENTS FIGURE 5-12-7



Linear Misalignment



Angular Misalignment

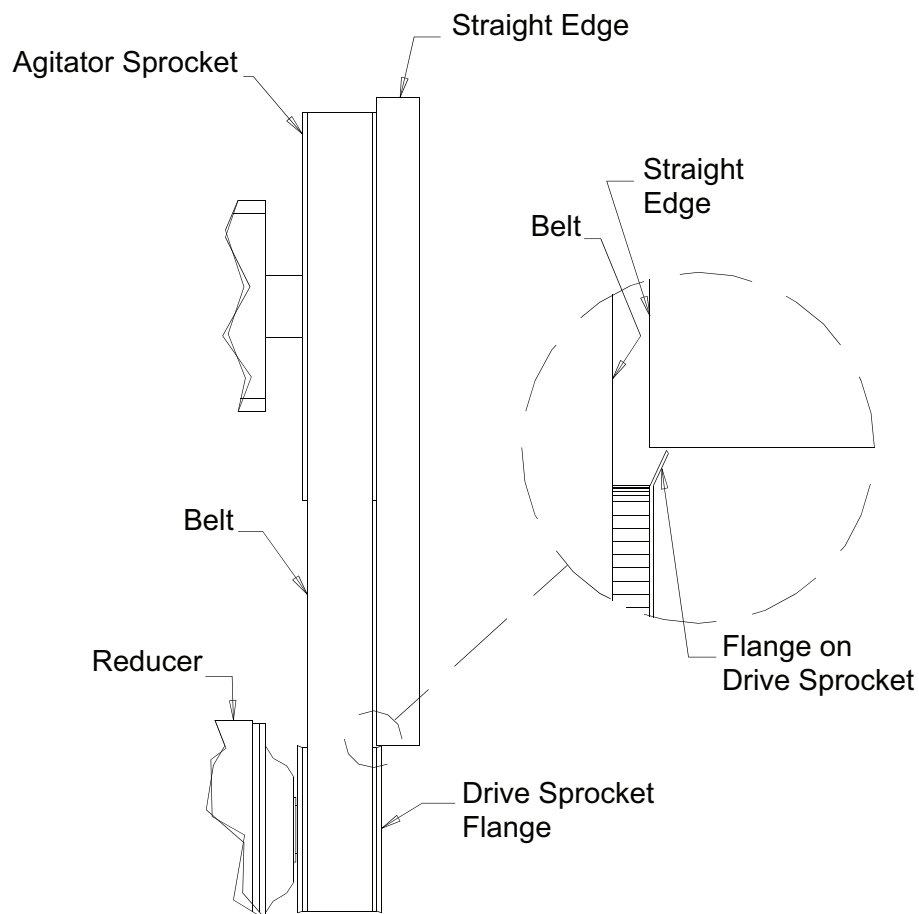
SECTION 5-12

TO CHECK LINEAR BELT ALIGNMENT:

1. Hold one end of a straight-edge across the face of the agitator sprocket so it lies against top and bottom edges. See **Figure 5-12-8, Linear Belt Drive Alignment**.
2. Slide the other end of the straight-edge down to the flange on the drive sprocket. Move the sprockets in or out until the edge of the straight-edge that is against the agitator sprocket is even with the inside edge of the flange on the drive sprocket.

NOTE: Set screws on the drive sprocket are usually doubled. Remove top screw and loosen bottom screw to move drive sprocket.

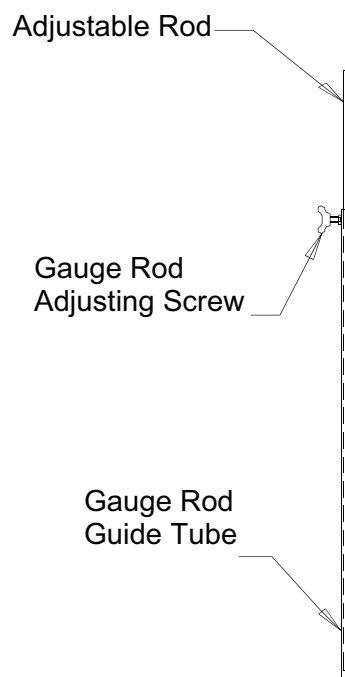
LINEAR BELT DRIVE ALIGNMENT FIGURE 5-12-8



TO CHECK ANGULAR BELT ALIGNMENT:

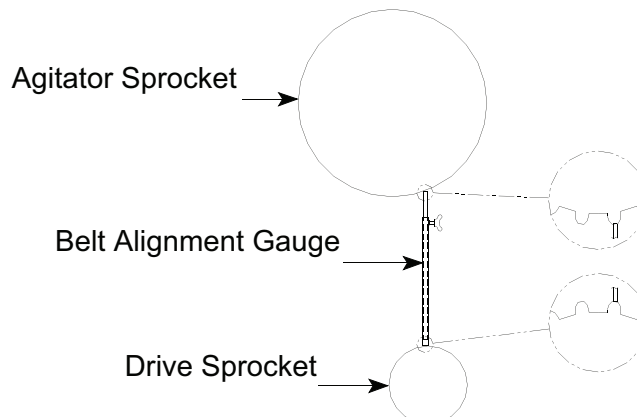
Use a **Belt Alignment Gauge**, supplied beginning with all 1997 orders. If you need and do not have a **Belt Alignment Gauge**, call the Parts Department at **Peerless Machinery Corp.** and order Part Number HS10-11-C64. Or you can, instead, make a gauge per **Figure 5-12-9**.

BELT ALIGNMENT GAUGE FIGURE 5-12-9



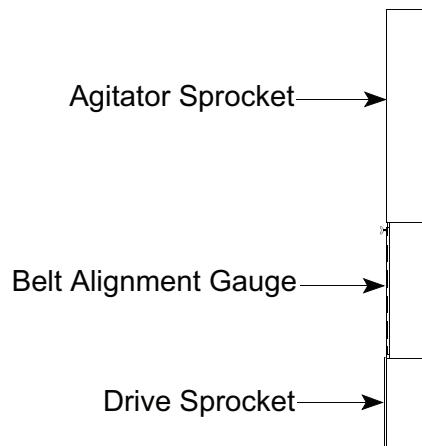
Follow the steps and illustrations below to check angular belt drive alignment:

1. Place **Belt Alignment Gauge** vertically from tooth to tooth on front edge of agitator and drive sprockets.

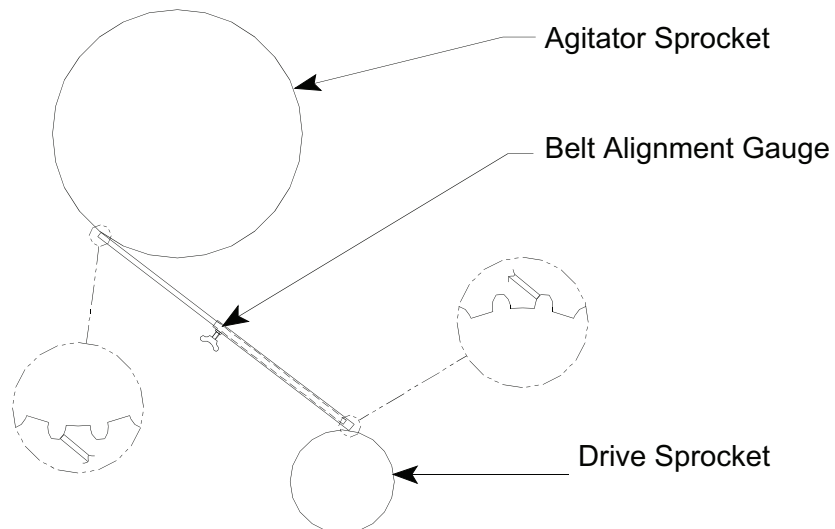


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2. Repeat step 1 at rear edge of sprockets.
3. Compare measurements using a feeler gauge to determine differences. For proper alignment, there must be no more than .010" difference with belt tight, or .005" with belt loose.



4. Place **Belt Alignment Gauge** as nearly horizontal as possible from tooth to tooth on front edge of agitator and drive sprockets.



5. Repeat step 4 at rear edge of sprockets.
6. Compare measurements using a feeler gauge to determine differences. For proper alignment, there must be no more than .010" difference with belt tight, or .005" with belt loose. See step 3 for illustration.

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If a misalignment of the **drive belt** system is found, check for these problems and make adjustments accordingly:

- ✓ The **gear reducer** and **drive motor** may need to be moved sideways.
- ✓ The **gear reducer** and **drive motor** may need to be reshimmed.

When making any of the above adjustments, remove the tension from the belt(s).

TO REPLACE DRIVE BELTS:

Peerless installs a single **drive belt** on mixers with **drive motors** up to and including 60 HP. Typically, any mixer with a rating of 75 HP and above requires two side-by-side **drive belts**.

If it becomes necessary to change the **drive belts** on a side-by-side belt system, *always change both belts*. Never change only one belt. Also be sure to specify when ordering that the belts *MUST* be a matched set, i.e., one belt split spanwise into two.

Use the following procedure to change the **drive belt(s)**:

1. Disconnect and lock out power to mixer. See **Section 1-3, Zero Energy State Procedures**.
2. Handle the **belt** to get a feel for its tension.
3. Loosen the 8 bolts on idler mounting bracket. Loosen jam nut that holds adjusting rod in place.
4. Slip **belt(s)** off agitator sprocket, then off drive sprocket.
5. Install new **belt(s)**. When a two **belt** set is used, make sure the **belt** logos face the same direction.
6. Retension idler so new **belts** feel like they have the same tension as before.
7. Run for 10 minutes under no load, and retension **belt(s)**.
8. After one day of operation, retension **belt(s)**.

SECTION 5-12**DRIVE MOTOR 5-12-3**

The **drive motor** supplies the power to rotate the mixer's agitator. **Drive motor** RPM is reduced by the **gear reducer** and the **drive chain/belt** and **sprockets** to the final agitator speed.

During scheduled maintenance, check the motor housing. Feel for unusual vibration; listen for unusual noise. Noise and/or vibration are signs of bearing or coupling failure. If problems occur, have a qualified motor service shop inspect the motor.

Excessive heat may be caused by an overloaded or dirty motor. In general, **drive motor** temperatures should not exceed 320° F (160° C) in the windings, and should not exceed 190° F (88° C) on outside housing when the mixer is running. Overloading or mixing a product with too low an absorption rate for the design of the mixer can cause the motor to overheat.

Insufficient voltage supply to the mixer can overheat the motor. See **Section 5-13, Electrical System**, for more details about **drive motor** overloading.

All motors are supplied with a 6-month supply of grease in the **drive motor** bearings. We strongly recommend greasing the **drive motor** every six months to prolong the life of the bearings. See **Table 5-2-5, Recommended Lubricants Chart** for type of grease to use. *Excessive or too frequent lubrication can damage the **drive motor**.*

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Use the following guide for greasing the **drive motor**.

⚠ DANGER

Before servicing, repairing, or cleaning mixer, follow **LockOut/TagOut** procedure as required by **OSHA Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

TO GREASE MOTOR ON SHAFT END:

On motors that come with pipe plugs only, remove the plug(s) on one side and install a grease zerk. Unscrew and remove the pipe plug on the opposite side of all motors for grease relief. Be sure that all fittings are free of dirt and otherwise clean. Using a standard low-pressure, hand-operated grease gun, pump 10 pumps of new grease in through the grease zerk. Run the motor for 3 minutes before replacing the pipe plug. See **Table 5-2-5, Recommended Lubricants Chart**.

TO GREASE MOTOR ON OPPOSITE DRIVE END:

Follow the procedure for the drive end except that the grease relief pipe extending through the fan cover acts as a reservoir for purged grease. Remove fan cover only if no grease fittings extend through the cover.

For motors that do not have pipe plugs or grease fittings in bearing housings, lubricate by removing the end shields from the motor. Clean the grease cavity and refill three-quarters full with the recommended grease. See **Table 5-2-5, Recommended Lubricants Chart**.

DRIVE MOTOR MAINTENANCE

Have all **drive motor** maintenance, other than lubrication, performed by an authorized motor service shop only.

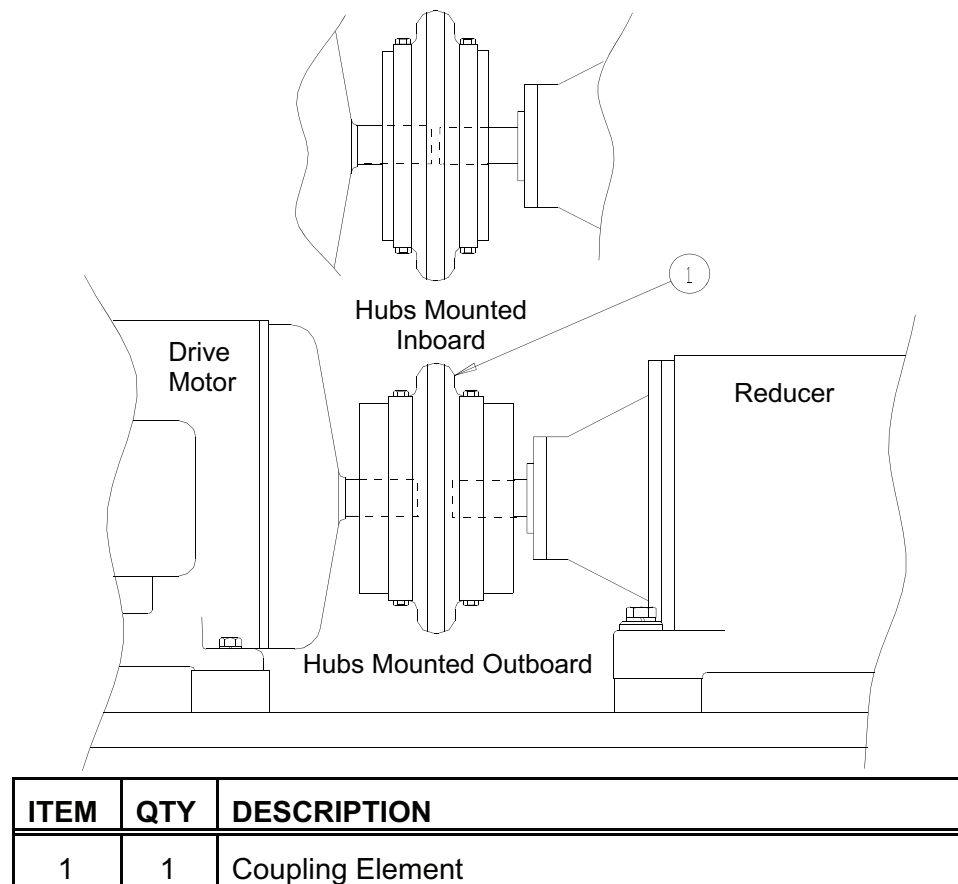
SECTION 5-12

MOTOR COUPLING 5-12-4

The **motor coupling** links the mixer's **drive motor** to the **gear reducer**. The coupling consists of the **drive motor** shaft, **gear reducer** shaft, hubs, and two rubber elements. The rubber elements are designed to absorb the shock loading and inconsistencies of the two rotating shafts. See **Figure 5-12-11, Drive Motor Coupling**.

Inspect the motor coupling for cracks or damage to the rubber element. If replacement of the rubber element is necessary, *always replace both half-elements*. Replacing only one half-element can cause torsional and balance problems.

DRIVE MOTOR COUPLING FIGURE 5-12-10



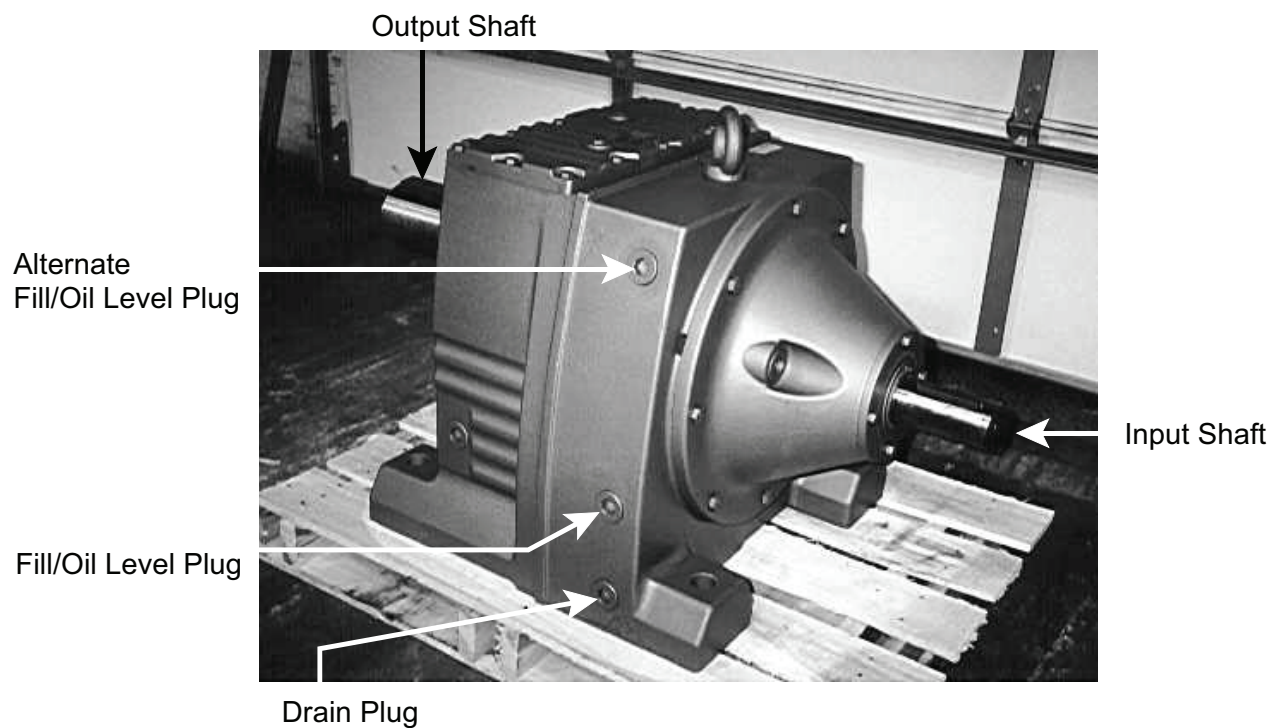
SECTION 5-12

GEAR REDUCER 5-12-5

The **gear reducer** is the first component in the process of reducing RPMs from **drive motor** speed to agitator speed. **Peerless** fits all of its Double Sigma Arm mixers with Eurodrive **gear reducers** as standard.

Eurodrive **gear reducers** have a top-mounted vent plug to allow for pressure release. A drain plug and a fill/oil level plug are located on the side of the unit that faces the back of the mixer so they can be easily reached for servicing. See **Figure 5-12-11, Eurodrive Gear Reducer**.

EURODRIVE GEAR REDUCER FIGURE 5-12-11



NOTE: Normal **gear reducer** operating temperatures should not exceed 180° F on the surface of the housing or 200° F internal oil temperature.

TO CHECK OIL LEVEL IN A EURODRIVE REDUCER:

Remove the fill plug. Use one or both of the following methods to check the oil level:

- ✓ Look inside the fill-plug opening. If oil oozes out or you can see it through the opening, the lubrication level is sufficient. If not, add oil.
- ✓ Put a finger inside the fill plug opening and reach toward the end of the plug-opening threads. If you can feel oil, the lubrication level is sufficient. If not, add oil.

When checking oil level, inspect for burnt, discolored, or foamy oil. Also check the oil for metal particles. Any of these conditions could indicate a problem with the **gear reducer**. Contact the Service Department at **Peerless Machinery Corp.**, or a qualified local repair shop, for assistance.

TO ADD OIL TO A EURODRIVE REDUCER:

- ✓ Be sure the drain plug is tightly closed.
- ✓ Remove the vent plug or fill plug, and add oil through either.
- ✓ Check the fill plug opening. When oil is visible through the opening, the lubrication level is sufficient. See **Table 5-12-12** for oil capacity information.

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OIL CAPACITIES FOR EURODRIVE GEAR REDUCERS TABLE 5-12-12

Model	Capacity U.S. Gals. (liters)
R87	.61 (2.3)
R97	1.2 (4.6)
R107	1.6 (6.0)
R137	2.4 (9.0)
R147	4.1 (15.4)
R167	7.1 (27.0)

RIGHT-HAND GEAR CASE ASSEMBLY 5-12-6

⚠ DANGER

Rotating parts can cause serious injury or death. Disconnect and lockout electrical power before servicing, repairing or otherwise working on mixer. See **Section 1-3, Zero Energy State Procedures**.

⚠ WARNING

Gear case oil is flammable and can cause fire. Clean up immediately any gear case oil that may have leaked or spilled.

⚠ WARNING

Gear case oil is slippery and can cause falls. Clean up immediately any gear case oil that may have leaked or spilled as a result of maintenance.

The **right-hand gear case assembly** is located at the right-hand end of the mixing bowl and extends into the end column. The **right-hand gear case assembly** contains the gears and bearings to drive the two agitator shafts. The gears are immersed in an oil bath inside the **right-hand gear case assembly**. See **Figure 5-12-13, Right-Hand Gear Case Assembly**.

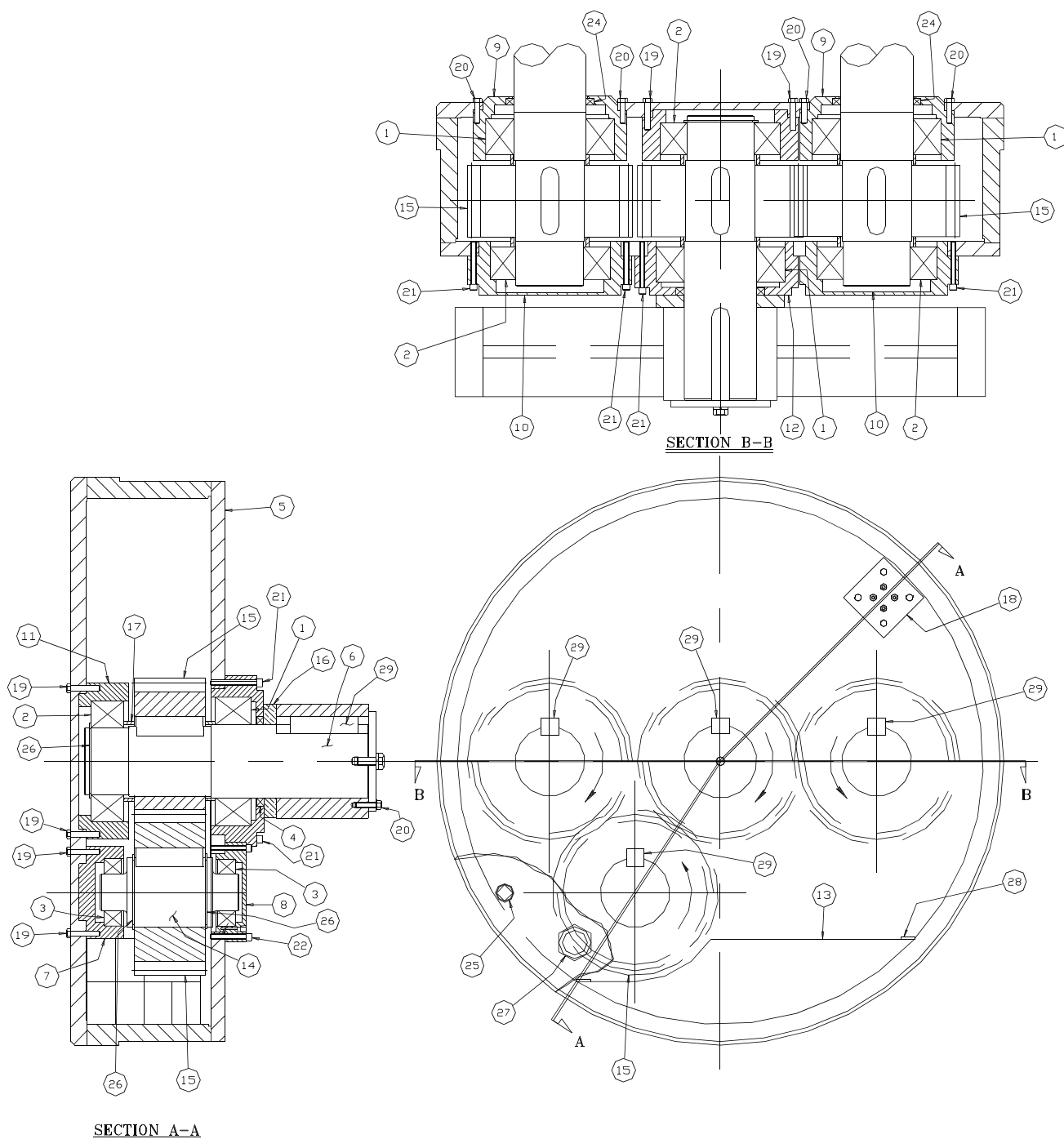
SECTION 5-12

At mixer installation and after one week of operation, perform the following preventive maintenance inspections. Thereafter, we recommend making these inspections as follows:

- ✓ Listen to right-hand gear case assembly running under load for any unusual noise or vibration. Also check for hotter than normal temperatures of bearings, shafts, and covers. If there are problems, call the Service Department at **Peerless Machinery Corp.** Perform weekly.
- ✓ Remove magnetic oil drain plug. Check oil quality and remove any metal shavings adhering to magnetic drain plug. Check the oil level in the gear case by looking at the oil level sight glass with the bowl in the full-up position. Perform monthly.
- ✓ Retorque all bolts. Torque 3/8" socket head bolts to 45 ft-lbs (61.02 Nm). Perform monthly.
- ✓ Drain existing oil; flush and refill gear case. When full, the oil level will be visible in the center of the oil sight gauge with the bowl in the full-up position. Perform every six months.

SECTION 5-12

RIGHT-HAND GEAR CASE ASSEMBLY FIGURE 5-12-13



SECTION 5-12

RIGHT-HAND GEAR CASE ASSEMBLY FIGURE 5-12-13

(con't.)

ITEM	QTY	DESCRIPTION
1	3	Spherical Roller Bearing
2	3	Spherical Roller Bearing
3	2	Spherical Roller Bearing
4	1	JM Split Clipper Seal
5	1	Gear Case Cover Assembly
6	1	Drive Shaft
7	1	Inboard Idler Bearing Housing
8	1	Outboard Idler Bearing Housing
9	2	Inboard Shaft Bearing Housing
10	2	Outboard Shaft Bearing Housing
11	1	Inboard Drive Bearing Housing
12	1	Outboard Drive Bearing Housing
13	3	Gear Case Filler
14	1	Idler Shaft
15	4	Gear
16	1	Bronze Spacer
17	6	Gear Spacer
18	1	Lube Lines Mounting Bracket
19	10	Hex Head Cap Screw S/S
20	12	Hex Head Cap Screw S/S
21	18	Socket Head Cap Screw M/S
22	4	Socket Head Cap Screw M/S
23	4	Hex Head Cap Screw M/S
24	2	C/R Oil Seal
25	1	Magnetic Pipe Plug
26	3	External Snap Ring
27	1	Oil Sight Gauge
28	2	Mounting Strip
29	4	Square Key

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SECTION 5-13

ELECTRICAL SYSTEM

⚠ DANGER

High electrical voltage is present in electrical enclosure. Only a qualified, experienced electrician should open mixer panel door.

⚠ DANGER

Before servicing, repairing, or otherwise working on mixer, follow **LockOut/TagOut** procedure as required by **OSHA Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

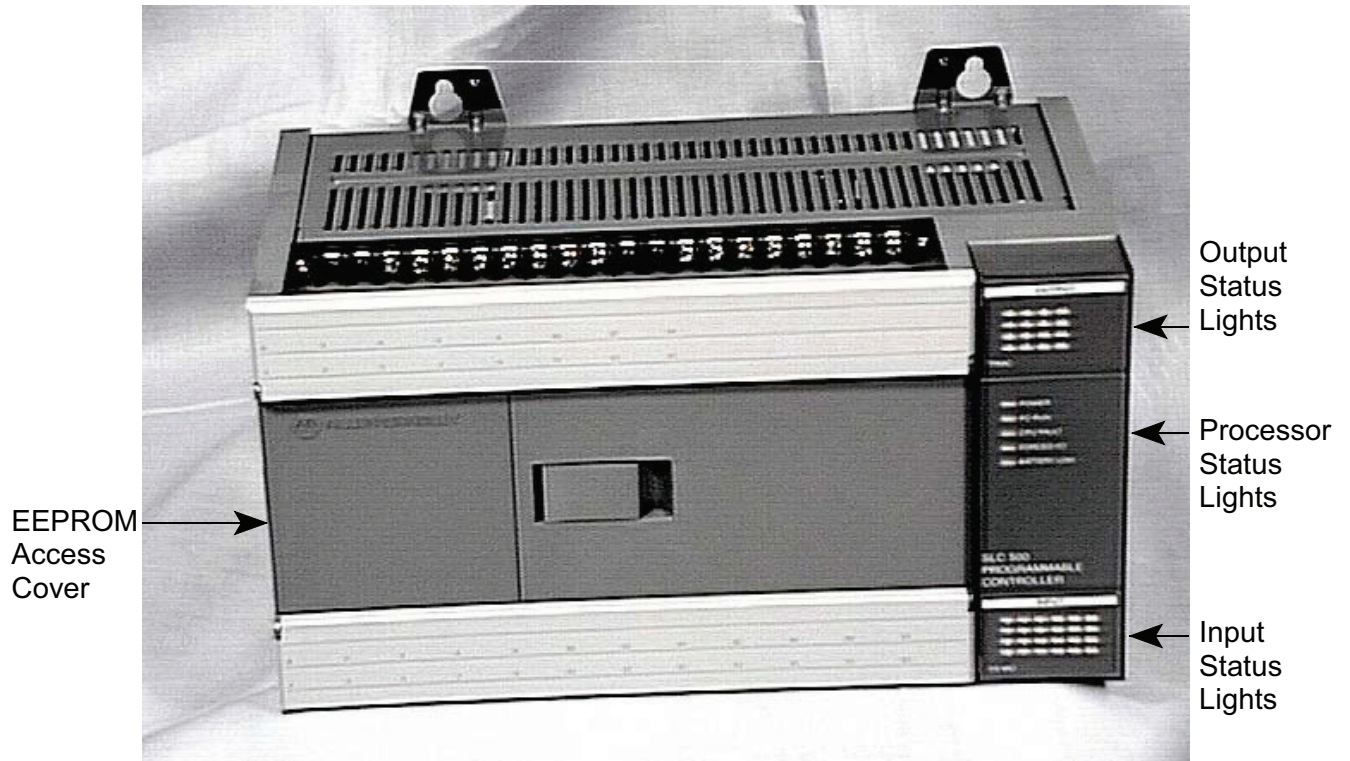
PROGRAMMABLE LOGIC CONTROLLER (PLC) 5-13-1

Your **Peerless** mixer is supplied with a **Programmable Logic Controller (PLC)**. The standard **PLC** used is Allen-Bradley model SLC 500. Many other types of **PLCs** may be used due to special mixer features or per customer request. See **Chapter 2, Options Checklist**, for the type of **PLC** installed on your mixer.

If your mixer has a special **PLC**, most of the following information still applies. In addition, the vendor's instruction manual for your **PLC** and related hardware is shipped with the mixer in the electrical enclosure.

SECTION 5-13

ALLEN-BRADLEY MODEL SLC 500 PLC FIGURE 5-13-1



PLC APPLICATION PROGRAM

⚠ DANGER

The **PLC** application program written by **Peerless** and supplied with your mixer is designed with all related safeties and interlocks. Do not modify program without prior written approval from **Peerless**. Making unauthorized changes to program may cause serious personal injury or death. In addition, machine damage and voiding of all warranties could result.

The application program is supplied by **Peerless**, loaded into the **PLC**, and tested prior to shipment in most cases. If **Peerless** does not supply the application program, special arrangements will be made for testing between the customer and **Peerless**.

SECTION 5-13

Peerless makes several options available concerning your application program:

- ✓ We will include a hard copy upon request.
- ✓ We can send copies on floppy disk per request.
- ✓ You can download the program from our Bulletin Board. Call the Engineering Department at **Peerless Machinery Corp.** for further assistance.
- ✓ We can send the program to your email address.

PLC HARDWARE

All **PLCs** have built-in diagnostic indicator lights to assist in trouble-shooting. The following list explains functions of the LED indicators on the standard Allen-Bradley Model SLC 500 fixed-style **PLC** used. See **Figure 5-13-1, Allen-Bradley Model SLC 500 PLC**.

- | | |
|---------------------|--|
| POWER. | ✓ Indicates PLC power is on. |
| PC RUN. | ✓ Indicates system check is complete and no problems have been found.
✓ System is ready to run. |
| CPU FAULT. | ✓ Indicates problem with the PLC . If lit, cycle power to the mixer. If indicator light does not go off, call local vendor or Peerless Service Department.
✓ When system is first powered up, CPU FAULT also lights but goes off after system check is complete. |
| FORCED I/O. | ✓ Indicates there is a forced input and/or output present. |
| BATTERY LOW. | ✓ Batteries are not used. See EEPROM Module below. |

SECTION 5-13**EEPROM MODULE**

The Allen-Bradley SLC 500 **PLC** uses an **Electrically Erasable Programmable Read Only Memory (EEPROM)** module to store, back up, or transfer SLC 500 application program. The SLC 500 can read and write to an **EEPROM**. Once it is plugged into the **PLC**, the **EEPROM** program overrides the application program in the CPU every time the power is cycled on, eliminating the necessity for a battery backup.

NOTE: Mixers with video control panels (**VCPs**) do not download the program from the **EEPROM** to processor memory when power is cycled. Downloading from the **EEPROM** would override any user-stored cycle values in the processor. For these applications, the program is downloaded on **CPU FAULT**.

If it becomes necessary to change the mixer's program, contact the Engineering Department at **Peerless Machinery Corp.** to request a revised **EEPROM** module. To change **EEPROM** modules, follow the procedure below. If the mixer has a **VCP**, you will need a means to download the new **EEPROM**.

TO REPLACE AN EEPROM MODULE:

To avoid damaging the **EEPROM** module, always handle by the ends or sides of the plastic housing. Keep in mind that skin oil and dirt can corrode metallic surfaces, inhibiting electrical contact. Do not expose the **EEPROM** module to magnetic surfaces, which can alter or delete memory.

Use the following procedure to install an **EEPROM** module. See **Figure 5-13-1, Allen-Bradley Model SLC 500 PLC**.

1. Disconnect all power to the mixer/**PLC** by turning off **MAIN POWER DISCONNECT** switch. See **Section 1-3, Zero Energy State Procedures**.
2. Remove the processor compartment cover.

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3. Grasp the carrier tabs with the thumb and index fingers, then gently but firmly lift upward on either end of the **EEPROM** module.
4. When the end of the module has come partially out of the socket, lift the other end the same way. Work both ends upward until the module is completely out of the socket.
5. Place the new **EEPROM** module correctly over the socket and press firmly in place. The **EEPROM** is keyed, so it will only go into the socket one way.
6. Replace the cover on the SLC 500 controller.
7. Restore power.
8. Test all functions of the mixer. Take special care to test all safety functions.

ELECTRICAL SYSTEM TROUBLE-SHOOTING GUIDE 5-13-2

Should problems occur with the operation of your mixer, *have only a qualified electrician check the I/O status indicators on the PLC*. In general, an **input** indicator lights when the device connected to it is functioning properly. If the device connected to the **input** is not operating, the indicator will not light.

The same holds true for **outputs**. If the status light is lit for a particular **output**, the device connected to that **output** should be functioning. However, if the **output** indicator is lit but the device is not operating, check for problems between the **PLC** and the device.

SECTION 5-13**MOTOR THERMAL OVERLOAD MODULE (MTOM) 5-13-3**

The **motor thermal overload module (MTOM)** monitors the drive motor temperature to protect the drive motor from excessive heat damage.

The **MTOM** module is mounted inside the electrical enclosure. Two thermistor wires are run from the **MTOM** to the three thermistors wired in series which are installed in the motor windings. Extreme motor temperature and/or a resistance of 20,000 Ω will cause the **MTOM** to trip, which will stop the mixer. After the drive motor has cooled, the **MTOM** can be reset by turning the power to the mixer off, then back on.

Some reasons the **MTOM** will trip are:

- | | |
|----------------------------|-------------------------------|
| ✓ Locked rotor | ✓ High motor ambient |
| ✓ Running overload | ✓ Blocked ventilation |
| ✓ Single phasing | ✓ High altitude conditions |
| ✓ Voltage unbalance | ✓ Fault in thermistors |
| ✓ Very dirty motor housing | ✓ Fault in thermistor wiring. |

Check the thermistors by measuring the resistance through them. Follow these steps:

1. Disconnect and lock out power to the mixer. **See Section 1-3, Zero State Energy Procedures.**
2. Disconnect the wires from the thermistors at the terminal strip in the electrical enclosure.
3. Using an **OHM** meter, measure resistance through the wires.
4. With the motor at room temperature, resistance should be less than 7500 Ω . If the resistance is greater than 7500 Ω , or if an open circuit is measured, check wiring to the motor.
5. If the motor wiring checks out, consult a certified motor shop for further testing.
6. Reconnect the wires to the terminal strip.

SECTION 5-13**OVERLOADING 5-13-4**

Mixer **drive motor** overloading can cause reduced mixer life and unscheduled down time. Common signs of mixer overloading are:

- ✓ Motor starter overload relay tripping (See **Section 5-13-5.**)
- ✓ Circuit breaker tripping
- ✓ **MTOM** tripping
- ✓ Motor load meter staying peaked for long periods of time.

If the mixer drive motor shows signs of overloading, the following solutions have proven helpful. Consult the Service Department at **Peerless Machinery Corp.** for assistance.

- ✓ Reduce amount of product being mixed
- ✓ Slow down agitator speed
- ✓ Increase agitator bars-to-bowl sheet spacing
- ✓ Remove breaker bar.

SECTION 5-13**STARTER OVERLOAD RELAY 5-13-5**

Peerless uses a solid-state **starter overload relay** for protection against drive motor overload. The overload relay is located at the bottom of the drive starter. The drive motor wires are connected to the relay's terminals.

The main advantage of the solid-state overload relay over the typical eutectic alloy overload relay is its response to a phase loss condition. A typical overload relay can take up to 40 seconds to trip when a motor is single phasing. The solid-state overload relay is designed to trip in 2 seconds preventing the possibility of serious damage to the mixer drive motor.

The solid-state overload relay will also stop the drive motor if motor overload occurs. The time it takes the overload relay to trip and shut off an overloaded drive motor varies with the magnitude of the motor overload. For example, a motor overload of 150% will take about 60 seconds to trip the overload relay. An overload of 200% will take about 11 seconds.

When the overload relay trips, a yellow display shows on the front of the overload relay indicating it has been tripped.

To reset a tripped overload relay, push the white *RESET* lever on the front of the overload relay.

NOTE: The trip selector setting on the overload relay must be set to the **Man** position.

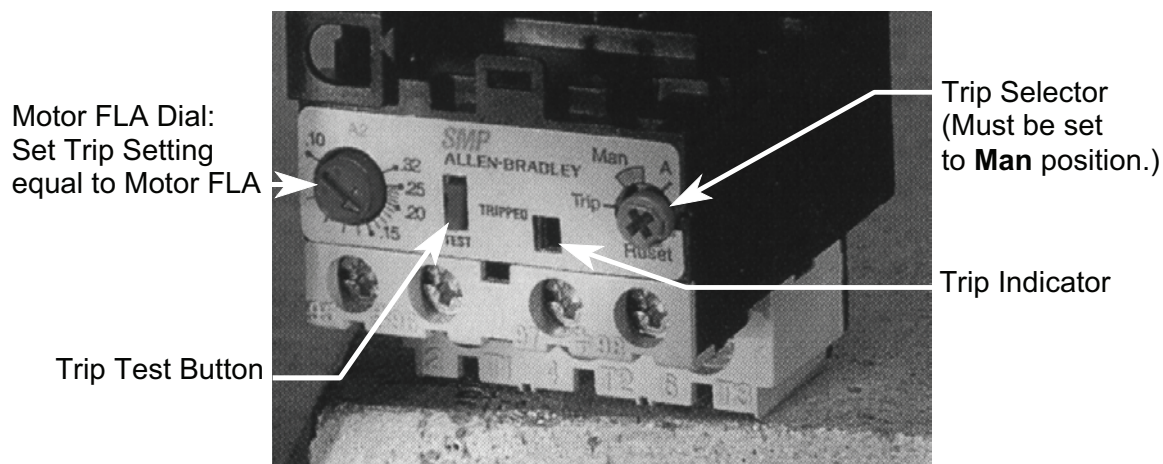
Unlike conventional eutectic alloy overload relays which use different elements or heaters to calibrate the relay to the motor Full-Load-Amps (FLA), the solid-state overload relay uses an adjustable dial or pot. The dial is factory set to the FLA of the drive motor. If the dial is adjusted to a higher amperage setting, severe damage to the drive motor can occur.

 WARNING

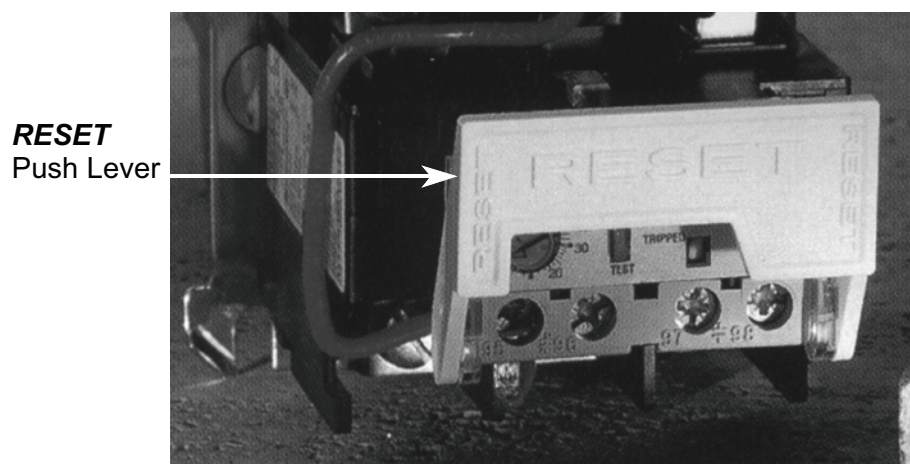
Do not adjust settings without written approval from **Peerless Machinery Corp.**

SECTION 5-13

STARTER OVERLOAD RELAY FIGURE 5-13-2



STARTER OVERLOAD RELAY WITH *RESET* PUSH LEVER FIGURE 5-13-3



SECTION 5-13**CUTLER-HAMMER IDT PANELMATE® 3000 VCP REPAIR 5-13-6**

Should your **Cutler-Hammer IDT PanelMate® 3000 VCP** fail, use the following procedures or call the Engineering Department at **Peerless Machinery Corp.** You can also call your local Cutler-Hammer IDT representative.

If problems occur with the **Cutler-Hammer IDT PanelMate® 3000 VCP** unit, consider the following options:

- ✓ Replace complete **VCP** unit
- ✓ Replace Central Processing Unit (**CPU**) only
- ✓ Replace front screen and bezel
- ✓ Replace backlight unit.

The complete unit or failed component can be sent to either **Peerless Machinery Corp.** or Cutler-Hammer IDT for repair or replacement.

A very dim screen is an indication that the **VCP's** backlight unit has failed.

Any **VCP** unit supplied as a replacement must be programmed with the mixer application program. If **Peerless** supplies the replacement unit, the application program will have been downloaded to the **VCP** before shipment.

NOTE: A Cutler-Hammer IDT PanelMate® 3000 VCP unit containing downloaded mixer application program can be stored for an indefinite period of time.
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SECTION 5-13

If **Peerless** is not the unit supplier, three options are available for programming:

- ✓ Send **VCP** unit to **Peerless Machinery Corp.**, Attn: Engineering Department. Please call first to make arrangements.
- ✓ If you have the required hardware and software, you can download the program.
- ✓ Make arrangements to have your local Cutler-Hammer IDT representative download the program.

If you or your Cutler-Hammer IDT representative intend to program the **VCP** but do not have the application program, call the Engineering Department at **Peerless Machinery Corp.** for assistance:

- ✓ We can send copies on floppy disk per request.
- ✓ You can download the program from our Bulletin Board. Call the Engineering Department at **Peerless Machinery Corp.** for further assistance.
- ✓ We can send the program to your email address.

In order to program the replacement **VCP**, you need the following:

- | | |
|--|--|
| ✓ Laptop computer: | <ul style="list-style-type: none">• 386 or higher• Serial port• MS Windows 3.1 or greater. |
| ✓ Cutler-Hammer IDT programming software: | Model No. 0620 |
| ✓ Cutler-Hammer IDT PLC Driver:
(Allen-Bradley SLC 500 line only) | Part No. 0741 |
| ✓ Transfer cable: | Part No. 0518 |
| ✓ Application program: | Call Peerless Machinery Corp. |

SECTION 5-13**TO REPLACE COMPLETE VCP UNIT:**

1. Disconnect all power to mixer by turning off **MAIN POWER DISCONNECT** switch. See **Section 1-3, Zero Energy State Procedures**.
2. Open mixer end doors and remove **VCP** access cover located behind **VCP** unit.
3. Disconnect power wires from **VCP** by removing black AC power cover on side of **CPU** and disconnecting G-N-L terminals.
4. Unplug PLC communication cable from Serial Port 2.
5. Unscrew (16) nuts from **VCP** mounting screws on back of bezel to detach from mixer.
6. Lift **VCP** unit from front of mixer.
7. Set in new **VCP** unit and screw (16) nuts onto **VCP** mounting screws to mount on mixer.

NOTE: **VCP** must be programmed with application program. See **Section 5-13-5**.

8. Plug PLC communication cable back into Serial Port 2.
9. Restore power wires to **VCP** by reconnecting G-N-L terminals. Replace AC power cover.
10. Replace access cover.
11. Restore power to mixer.

TO REPLACE FRONT SCREEN AND BEZEL:

1. Disconnect all power to mixer by turning off **MAIN POWER DISCONNECT** switch. See **Section 1-3, Zero Energy State Procedures**.
2. Open mixer end doors and remove **VCP** access cover located behind **VCP** unit.
3. Disconnect power wires from **VCP** by removing black AC power cover on side of **CPU** and disconnecting G-N-L terminals.
4. Unplug PLC communication cable from Serial Port 2.

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5. Unscrew (16) nuts from **VCP** mounting screws on back of bezel to detach unit from mixer.
6. Lift **VCP** unit from mixer.
7. Unplug ribbon cable connector from side of CPU.
8. Unscrew (2) CPU mounting screws located on top of unit and lift unit from bezel.
9. Place **CPU** unit on new bezel and screw in (2) CPU mounting screws.
10. Plug in ribbon cable connector on side of CPU.
11. Remove (16) rubber caps from mounting screws on new bezel.
12. Set in new **VCP** unit and screw (16) nuts onto **VCP** mounting screws on back of new bezel to reattach unit to mixer.
13. Plug PLC communication cable back into Serial Port 2.
14. Restore power wires to **VCP** by reconnecting G-N-L terminals. Replace AC power cover.
15. Replace access cover.
16. Restore power to mixer.

TO REPLACE CPU ONLY:

1. Disconnect all power to mixer by turning off **MAIN POWER DISCONNECT** switch. See **Section 1-3, Zero Energy State Procedures**.
2. Open mixer end doors and remove **VCP** access cover located behind **VCP** unit.
3. Disconnect power wires from **VCP** by removing black AC power cover on side of **CPU** and disconnecting G-N-L terminals.
4. Unplug PLC communication cable from Serial Port 2.
5. Unscrew (16) nuts from **VCP** mounting screws on back of bezel to detach unit from mixer.
6. Lift **VCP** unit from mixer.
7. Unplug ribbon cable connector from side of **CPU**.
8. Unscrew (2) **CPU** mounting screws located on top of unit and lift unit from bezel.

SECTION 5-13

9. Place new **CPU** unit on bezel and screw in (2) **CPU** mounting screws.
10. Plug in ribbon cable connector on side of **CPU**.
11. Set in **VCP** unit and screw (16) nuts onto **VCP** mounting screws on back of new bezel to reattach unit to mixer.
12. Plug PLC communication cable back into Serial Port 2.

NOTE: **VCP** must be programmed with application program. See **Section 5-13-5**.

13. Restore power wires to **VCP** by reconnecting G-N-L terminals. Replace AC power cover.
14. Replace access cover.
15. Restore power to mixer.

NOTE: After repair or replacement of **VCP** or its components, be sure **VCP POWER** button on side of unit is in the ON position before restoring power to mixer.

TO REPLACE GRAYSCALE BACKLIGHT:

1. Disconnect all power to mixer by turning off **MAIN POWER DISCONNECT** switch. See **Section 1-3, Zero Energy State Procedures**.
2. Open mixer end doors and remove **VCP** access cover located behind **VCP** unit.
3. Disconnect power wires from **VCP** by removing black AC power cover on side of **CPU** and disconnecting G-N-L terminals.
4. Unplug PLC communication cable from Serial Port 2.
5. Unplug membrane keyboard ribbon cable.
6. Loosen (2) thumbscrews on top of **CPU**.
7. Lift **CPU** from back of bezel. Take to clean, static-free work bench.
8. Remove cover from **CPU**.

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9. Disconnect ribbon cable and power cable.
10. Remove all boards from inside **CPU**.
11. Disconnect circuit board from **CPU**.
12. Disconnect backlight cable.
13. Remove display module from **CPU**.
14. Remove the backlight cartridge from display module by pulling out white tab. (Backlight cartridge will snap out.)
15. Insert new backlight cartridge into display module. (Backlight cartridge will snap into place.)
16. Reconnect display module to **CPU**.
17. Reconnect the backlight cable.
18. Reconnect circuit board to display module.
19. Slide boards back into **CPU**.
20. Reconnect ribbon cable and power cable.
21. Replace **CPU** cover.
22. Place **CPU** back on bezel.
23. Tighten (2) thumbscrews on top of **CPU**.
24. Plug in membrane keyboard ribbon cable.
25. Plug PLC communication cable into Serial Port 2.
26. Reconnect G-N-L terminals and **VCP** power wires. Replace black AC power cover on side of **CPU**.
27. Replace **VCP** access cover.
28. Restore power to mixer.

SECTION 5-13**TO REPLACE COLOR BACKLIGHT:**

1. Disconnect all power to mixer by turning off **MAIN POWER DISCONNECT** switch. See **Section 1-3, Zero Energy State Procedures**.
2. Open mixer end doors and remove **VCP** access cover located behind **VCP** unit.
3. Disconnect power wires from **VCP** by removing black AC power cover on side of **CPU** and disconnecting G-N-L terminals.
4. Unplug PLC communication cable from Serial Port 2.
5. Unplug membrane keyboard ribbon cable.
6. Loosen (2) thumbscrews on top of **CPU**.
7. Lift **CPU** from back of bezel. Take to clean, static-free work bench.
8. Remove cover from **CPU**.
9. Disconnect ribbon cable and power cable.
10. Remove all boards from inside **CPU**.
11. Disconnect circuit board from **CPU**.
12. Disconnect backlight cable.
13. Remove display module from **CPU**.
14. Remove screw behind backlight. (You may need a jeweler's screwdriver.)
15. Slide backlight cartridge from **CPU**.
16. Insert new backlight cartridge into display module and secure with screw.
17. Reconnect display module to **CPU**.
18. Reconnect backlight cable.
19. Reconnect circuit board to display module.
20. Slide boards back into **CPU**.

SECTION 5-13

21. Reconnect ribbon cable and power cable.
22. Replace **CPU** cover.
23. Place **CPU** back on bezel.
24. Tighten (2) thumbscrews on top of **CPU**.
25. Plug in membrane keyboard ribbon cable.
26. Plug PLC communication cable into Serial Port 2.
27. Reconnect G-N-L terminals and **VCP** power wires. Replace black AC power cover on side of **CPU**.
28. Replace **VCP** access cover.
29. Restore power to mixer.

TO ORDER REPLACEMENT**CUTLER-HAMMER IDT PANELMATE® 3000 VCP UNIT OR COMPONENTS:**

To replace your **Cutler-Hammer IDT PanelMate® 3000 VCP** or any of its components, call Cutler-Hammer IDT or your local IDT representative. The following information is intended as a guide to make ordering parts easier.

The following information can be found on a sticker attached to your **VCP** unit. When ordering a replacement **Cutler-Hammer IDT PanelMate® 3000 VCP** unit or components, give this information to IDT personnel along with the name of the unit or part desired:

- ✓ Catalogue number (37PK, grayscale display or 39PK, color display)
- ✓ Part number
- ✓ Serial number.

With the above information, IDT personnel can determine any parts used on your particular **VCP** unit.

Your **Cutler-Hammer IDT PanelMate® 3000 VCP** unit is comprised of:

- ✓ Complete **Cutler-Hammer IDT PanelMate® 3000 VCP** unit
- ✓ **Cutler-Hammer IDT PanelMate® 3000 VCP** custom front screen and bezel
- ✓ Central processing unit (**CPU**)
- ✓ Backlight Kit.

SECTION 5-14

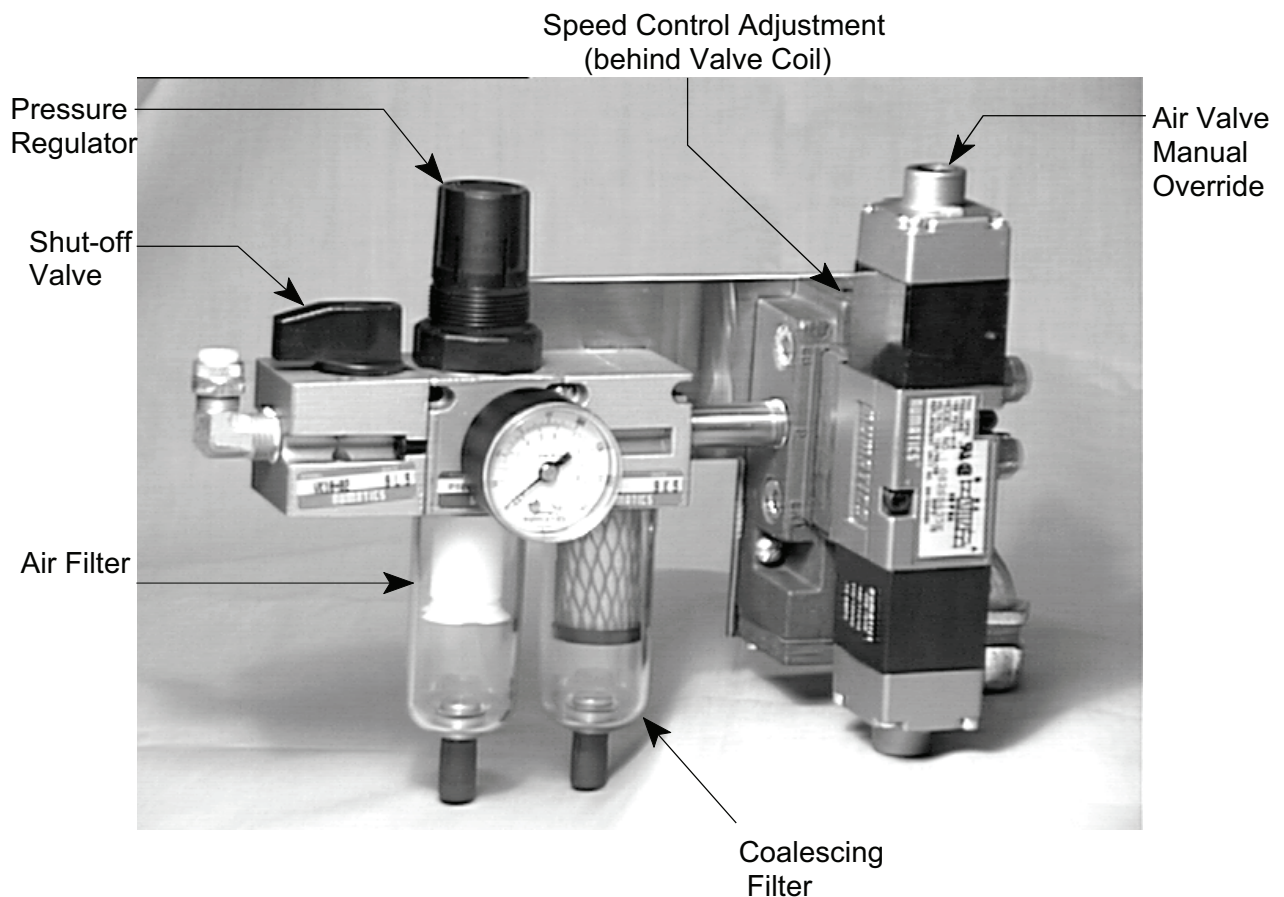
PNEUMATIC SYSTEM

⚠ CAUTION

Pressurized air can cause personal injury. Air pressure lines must be shut off, locked out and remaining air removed from system before working on mixer.

Typically, the **pneumatic (air) system** supplies pressurized air to operate the mixer's **flour gate**. When applicable, options like the **Auto Central Lube System**, **Trough Hooks**, etc. are also pneumatically powered. **Pneumatic System** components are located inside the end doors at the left-hand end of the mixer. See **Figure 5-14-1, Air Preparation Assembly**.

AIR PREPARATION ASSEMBLY FIGURE 5-14-1



SECTION 5-14

An **Air Preparation Assembly** is used on the mixer's air supply. The assembly consists of:

- | | | |
|----------------------------|---|--|
| Shut-off valve. | ✓ | Black ON/OFF valve shuts off air supply to mixer. |
| Air filter. | ✓ | Filters in-coming air. Replace when dirty, damaged, or when air pressure drop across filter slows gate movement. <i>Be sure to turn off air supply before changing filter.</i> |
| | ✓ | Drain bowl when water is present. Remove red vinyl cap at bottom and turn valve clockwise to raise drain plug until water begins to drain out. |
| Pressure regulator. | ✓ | Regulates air pressure to mixer. Adjust pressure by lifting and turning black knob on top of the regulator. |
| | ✓ | Gauge in front of regulator tells air pressure setting. |
| | ✓ | Set pressure at lowest setting that allows flour gate to function, usually 60-90 PSI. <i>Never exceed 100 PSI air pressure.</i> |
| Pressure switch. | ✓ | Prevents air system solenoids from energizing with no air present. |

NOTE: The pneumatic system on your **Peerless** mixer is designed to be operated without lubrication. ***Do not add lubricating oil to the Air Preparation Assembly.*** These components are not lubricators.

NOTE: Excess water in the air supply could damage internal parts of air system.

FOR SLIDE GATE OPTION ONLY:

The flour gate is controlled by an air valve. Two electric solenoids operate the air valve, which has a rear-mounted speed control unit for air flow adjustment to the air valve. The speed control contains two needle valves, one at the top and one at the bottom. Use a screwdriver to turn each needle valve, as desired, for independent adjustment of air flow.

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Adjusting the needle valves controls the operating speed of the flour gate. Screw *in* to slow down the speed of the flour gate. Screw *out* to speed up operation of the flour gate. Adjust one or both needle valves so the gate operates smoothly and quietly, and does not slam when opening or closing.

Peerless uses detent-type air valves which prevent accidental spool shift caused by machine vibration, shock, or loss of electrical power. The valve will shift only when the solenoid is energized. When the solenoid is de-energized, the valve will stay in its last position. This feature prevents accidental operation of air-powered components when electrical power is lost to the mixer.

The air valves can be manually cycled by depressing the recessed button on the end of the solenoid valve with a narrow-tipped tool (i.e., a screwdriver).

SECTION 5-15

INDIRECT REFRIGERATION SYSTEM

Peerless can supply your mixer with an **indirect refrigeration system** for maintaining a specific dough temperature during the mix cycle. See **Chapter 2, Options Checklist**, to find out if your mixer has been equipped with the optional **indirect refrigeration system**.

To ensure proper refrigeration system operation, a scheduled maintenance program is necessary. The following information is intended as a guideline. See **Chapter 7** for more information on refrigeration systems.

RELIEF VALVE

The refrigeration system is equipped with a **relief valve** to prevent over pressurization of the bowl coolant jacket. A **relief valve** prevents the bowl coolant jacket from exceeding 235 PSI. **Peerless** tests its bowl jackets at 250 PSI pressure.

The **relief valve** is located on the left-hand side of the bowl at the rear of the mixer. Check weekly for blockage.

HOSES

Breaking or cracking **hoses** indicates catastrophic failure could occur. The coolant system could lose its fluid. To reduce the chance of catastrophic failure, check **hoses** monthly for wear, damage and refrigerant leaks. Replace refrigeration **hoses** yearly whether or not failure occurs.

TO REPLACE COOLANT HOSES

1. Turn off valves at end of **hose**.
2. Crack fitting and drain **hose**.
3. Remove **hose**.
4. Install new **hose**.
5. Turn on valves.
6. Check for leaks.